

**Development of a Cross-Platform Ticketing and Task Management System for
the Municipal Information Technology Office in Montalban, Rizal**

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ABSTRACT

The study, titled “Development of a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal,” aimed to design, develop, and evaluate a system that simplifies request handling, task monitoring, and reporting processes within the Municipal Information Technology Office (MITO). The system was developed to address challenges in managing technical support requests, tracking task progress, and maintaining efficient communication between IT personnel and department heads.

Using the Spiral Model as the Software Development Life Cycle (SDLC), the researchers developed three platforms: a web application for department heads, a mobile application for municipal staff, and a desktop application for administrators. System evaluation was based on the ISO/IEC 25010 Software Quality Model, covering Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security.

Results showed that IT experts rated the system with a general weighted mean of 3.02 (Agree), indicating satisfactory compliance with quality standards, while respondents rated it 3.78 (Strongly Agree), confirming that the system fully meets software quality requirements.

Overall, the system effectively fulfills its objectives of improving efficiency in ticket management and task monitoring. The researchers recommend adding features such as a calendar of activities, an electronic signature, and enhanced connectivity to further optimize system performance.

Keywords: *Ticketing System, Task Management, Cross-Platform Application, Spiral Model, ISO/IEC 25010, Municipal Information Technology Office*

CHAPTER I

INTRODUCTION

This chapter includes the background of the study, objectives of the study, significance of the study, scope and delimitation, operational framework, and conceptual framework.

1.1 Background of the Study

In the early years of local governance, service request handling was primarily conducted through manual methods, including written logs, physical forms, and verbal communication. As information technology evolved, digital systems were introduced to simplify these processes, enabling organizations to handle requests more efficiently. However, many local offices, including the Municipal Information Technology Office (MITO) in Montalban, Rizal, continue to face challenges in fully adopting integrated digital systems for managing IT-related tasks and support requests.

In today's digital age, government offices are expected to work efficiently, transparently, and responsibly with the help of technology. As the use of IT services continues to grow, managing tasks and service requests in an organized way has become very important. Automated systems for tracking and handling requests help make operations faster and more reliable. Because of this, ticketing systems with task management features have become essential tools for improving coordination, accountability, and overall service quality in public offices.

A ticket refers to a recorded service request or issue submitted by a user that requires action, support, or resolution from the IT department. Each ticket includes important details such as the requester's information, problem description, priority level, and status updates. (*What Is an IT Ticketing System? - ServiceNow*, n.d.). A task, on the other hand, is a defined unit of work derived from a ticket or operational process that is assigned to a specific person or team. Tasks are prioritized, tracked, and completed to fulfill the requirements of the associated ticket. (*Task Management: Definition, Examples, Methods, Software*, n.d.). Together, these are managed through a

ticketing and task management system, a digital platform designed to automate the process of submitting, tracking, assigning, and resolving service requests within an organization. This system improves communication, accountability, and transparency by ensuring that every request is efficiently processed and documented.

The Municipal Information Technology Office (MITO) plays a crucial role in supporting the digital transformation and governance of the local government of Montalban, Rizal. Guided by its mission “to deliver effective management tools for efficient and transparent governance” and its vision “to systematize operations through automation and digitalization”, MITO ensures that all departments have reliable access to secure and well-maintained IT services. The office is responsible for installing, maintaining, and repairing hardware and software systems, as well as setting up workstations, configuring networks, and ensuring the security and privacy of digital infrastructures. In addition, MITO provides technical support, system development, and user orientation to enhance productivity and improve public service delivery. Upholding its core values of Mastery, Integrity, Technology-Driven solutions, and Organization, MITO remains committed to facilitating the technological needs of the municipality and optimizing performance across all government operations. (*Municipal Information Technology Office - Municipality of Montalban*, n.d.).

Previous studies have emphasized the importance of digital ticketing and task management systems in improving service delivery and operational efficiency. For instance, Abidemi (2024) reported that organizations implementing automated task management solutions achieved a 30% increase in efficiency and a significant reduction in service request backlogs. Similarly, Bhardwaj (2025) highlighted the advantages of cross-platform systems in enhancing accessibility and user engagement, enabling staff to manage tasks efficiently across web, mobile, and desktop interfaces. These studies collectively highlight how automation and digital integration improve responsiveness, accountability, and productivity within institutions.

According to Mr. Cornelio P. Luib, the former Officer-in-Charge of MITO and now the Officer-in-Charge of the Municipal Public Information Office (MPIO), the current service requesting process still relies on manual methods such as written logs,

emails, and verbal communication. This fragmented approach often results in delayed response times, a lack of task visibility, and inefficiencies in resource allocation. He further emphasized that service requests are sometimes lost or overlooked, while miscommunication among staff contributes to inconsistent resolution processes. Moreover, their existing open-source system is underutilized due to concerns over data privacy risks and functional limitations. In its current form, tickets are not properly isolated or categorized, and there is no effective monitoring of tasks.

To address these gaps, this study proposes the “Development of a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal”. This system is a cross-platform solution designed to simplify IT support processes and improve service efficiency. The proposed system automates the entire life cycle of municipal IT service requests from ticket submission and task assignment to progress tracking and completion, eliminating delays and inefficiencies brought by manual or fragmented methods. It features three main platforms: a web application, a mobile application, and a desktop application, each serving distinct roles in MITO’s operations. The web application is intended for department heads, allowing them to monitor service requests, assign tasks, oversee ticket progress, and generate reports for better decision-making. The mobile application is designed for municipal staff, enabling them to receive, update, and resolve tasks while on the field with real-time notifications and task updates, ensuring responsiveness and mobility. The desktop application serves as the admin's platform, providing centralized control over users, security, and system settings, while ensuring proper data management and accountability.

Each feature serves specific functions that collectively enhance MITO’s operations. Users could easily create and track tickets, could assign and monitor tasks with greater visibility, and the system provides real-time updates to keep all stakeholders informed. The system also generates reports and analytics, providing insights into ticket volume, response times, and staff performance, which help improve resource allocation and service planning. With its cross-platform accessibility, tasks could be managed efficiently across web, mobile, and desktop environments.

By combining these features and functions, the system directly addresses MITO's current challenges. Real-time ticketing reduces delays, the centralized dashboard improves task visibility, built-in notifications minimize miscommunication, and the controlled environment ensures better data privacy compared to their current underutilized open-source system. Moreover, mobile access allows staff to stay responsive even outside the office.

This study was initiated to address the growing need for an efficient, secure, and centralized system that supports MITO's goal of providing reliable IT services. By developing a cross-platform ticketing and task management system, the project aims to simplify workflows, improve response times, and enhance accountability across all levels of operation. The results of this study are expected to contribute to the modernization of MITO's service operations.

1.2 Statement of the Objective

General Objective

The main objective of this study is to develop and implement a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal, to enhance operational efficiency and improve the management of IT service requests.

Specific Objectives

Specifically, this study entitled “Development of a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal” aims to:

1. Develop a web application for department heads, which enables them to overview each department’s performance, perform CRUD (create, read, update, and delete) operations on tickets, task assignment, progress tracking, manage employees, and generate reports and analytics.
2. Develop a mobile application for municipal staff, enabling them to access and update assigned tasks in real time, attach files, comment, receive notifications, and coordinate efficiently through synchronized updates even while on the move.
3. Develop a desktop application for administrators that provides centralized control over ticket management, user management, oversees department management, system settings, offers analytics and security, complete monitoring of service requests, and generates reports.

General Features of the Ticketing and Task Management System

The proposed Cross-Platform Ticketing and Task Management System aims to improve the efficiency and organization of request handling at the Municipal Information Technology Office (MITO) in Montalban, Rizal. It consists of three integrated platforms: a web, mobile, and desktop application, each designed for department heads, municipal staff, and administrators. These platforms work together to provide smooth communication, real-time updates, and centralized system control. The key features of each platform are outlined below:

1. Web Application

The web application is designed to support department heads in overseeing their team's performance and managing tasks efficiently. Its features include:

i. Login/Sign Up Page

This is the initial entry point where a Department Head provides their Email Address and to securely access the system. A welcome message confirms the user type, and options are available for registration or password recovery.

ii. Dashboard / Main Hub

The Dashboard serves as the central control panel for the MITO system, providing an immediate and comprehensive summary of departmental operations and performance upon login. The user is greeted with real-time metrics for all Tickets and Tasks (Total, Open, Overdue), as well as Active Users. This hub also displays a detailed performance overview, highlighting team efficiency, completed work, and individual employee performance ratings.

iii. Tickets Management

This page is the department's dedicated area for handling all support requests and departmental tickets. It allows Department Heads to track, categorize, and act on tickets they have sent, received, or forwarded. A prominent button allows creating new tickets quickly, and the interface includes search and filtering tools based on Status and Priority to manage workload efficiently.

iv. Tasks Management

This module is used to create, organize, and track all departmental tasks and projects. A button is available for adding new tasks instantly. The main view presents a clean table of all tasks, showing critical details like Title, Priority, Status, Assigned To employee, and Due Date, with actions to view or edit tasks.

v. Kanban Board

The Kanban Board provides a dynamic, visual management tool where Department Heads can drag and drop tasks and tickets across workflow stages: Open / To Do, In Progress, and Done / Completed. A color-coding system helps differentiate between work related to the Internal (same) Department and External (other) Departments, facilitating a clear, collaborative, and agile overview of the department's workload.

vi. Department Analytics

This page offers comprehensive, visual insights into the team's performance and productivity. It converts operational data into actionable graphs that track monthly progress for tasks and tickets (categorized by status). The module also features a Team Performance Overview section, which is crucial for assessing efficiency and highlighting top performers.

vii. Reports Module

This is the system's area for generating and managing comprehensive reports. Department Heads could create new custom reports based on specific data (tickets, tasks, timeframes, or employee performance). The page includes filters to narrow the scope of generated reports and a centralized list of ALL REPORTS for easy review and retrieval of past analysis.

viii. Employees Module

The Employees module is the primary administrative tool for managing all personnel within the department. It displays a clean table of every employee, showing essential information such as their Name, Email, Role, Department, and Status. The page includes a search bar and actions for viewing details and performing administrative tasks on employee accounts.

ix. Settings Module

The Settings module is the central control panel for the user to manage application preferences, account details, and security options. It uses tabs to separate functionalities: Account (to view/edit personal details and department role) and Security (for managing passwords). This module ensures the user can easily maintain their profile and secure system access.

x. Notifications

This is the central place to track all real-time alerts and updates related to tickets and tasks. The screen displays a clear list of events, such as a comment on a task. Tabs are provided to quickly filter notifications by All, Unread, or Read status.

2. Mobile Application

The mobile app allows municipal staff to access and manage tasks on the go, ensuring efficient communication and real-time updates. Its features include:

i. Login/Sign Up Page

This is a secure entry screen where an Employee User accesses the mobile app. The user simply enters their Email and Password to authenticate. Options are included to register a new account or recover a lost Password.

ii. Mobile Dashboard

The Dashboard is the main screen after logging in, providing a quick snapshot of the employee's most important work. It uses simple cards to display critical counts like Open Tasks and any Overdue items. A Recent Activity feed keeps the user updated on new comments or events. All main sections (Home, Tasks, Notifications, Profile) are always accessible via the bottom menu bar.

iii. Mobile Task Details

This screen is for viewing and managing the details of a specific task. It clearly shows the task's title, its current Status (e.g., in progress), Priority (e.g., medium), who it is Assigned To, and the Due Date. The user could also edit the task details, Attach Files for documentation, and view or add Comments for team collaboration.

iv. Mobile Notifications

This is the central place to track all real-time alerts and updates related

to tickets and tasks. The screen displays a clear list of events, such as a comment on a task. Tabs are provided to quickly filter notifications by All, Unread, or Read status.

v. Mobile Profile/Account Settings

The Profile screen allows the user to view and manage personal account and security settings. It shows the user's name, email, and assigned Department. Important action buttons are included for the user to Edit Profile, Change Password, and secure Logout from the application.

3. Desktop Application

The desktop application provides administrators with full control over the system, ensuring security and operational efficiency. Its features include:

i. Desktop Login Page (Admin Access)

This screen is the secure entry point where an Administrator accesses the system. The user must provide their Email Address and Password to authenticate, with options available to retrieve a forgotten password or register a new account. A large "Sign In" button finalizes the process, granting access to the system's administrative functions.

ii. Desktop Dashboard (Admin View)

The Dashboard is the central control panel that provides a high-level, real-time overview of the system's operational health immediately upon login. It prominently displays key metrics such as Total Tickets, Total Tasks, Active Users, and managed Departments. The page also highlights System Performance metrics (like Task Completion Rate) and provides a Quick Actions section with direct links to management tasks like reports and user management.

iii. Desktop Analytics Dashboard (Admin View)

This is a data-focused page for the admin to gain comprehensive performance insights into the organization's productivity. Admins could filter the data by Department and Date Range, and the screen displays critical KPIs like Total Employees, Department Efficiency, and Average Task Completion time. Detailed graphical metrics for both Task and Ticket performance are provided below these KPIs for strategic decision-making.

iv. Desktop Reports Module (Admin View)

This screen is the dedicated area for the admin to generate, manage, and review all departmental and user reports. The primary action is the "+ New Report" button, and the reports are organized into tabs for "All Reports," "Recent," and "Scheduled". Admins could apply powerful filters by Report Type and Date Range to efficiently locate specific data sets, with generated reports visually showcased by subject and creation date.

v. Desktop Tickets Management

The Tickets Management screen serves as the primary hub for tracking and handling all support requests and departmental tickets. A prominent "+ New Ticket" button allows for the immediate creation of a new request. The interface is organized by three tabs: "Sent Tickets," "Received Tickets," and "Forwarded by Me," and filters are available to narrow the view by Search, Status, and Priority, ensuring efficient management of tickets.

vi. Desktop Users Management (Admin View)

This is the critical administrative interface used to oversee and manage all active user accounts and pending employee approvals. An Admin could quickly board new personnel using the "+ Add New User" button. The

section is divided into View Active Users (including Department Heads and Employees) and Pending Users, with user profiles displayed on cards that include the user's name, role, approval status, and a "View Details" option for in-depth profile management.

vii. Desktop Department Management (Admin View)

The Department Management screen is the dedicated area for an Admin user to organize, manage, and oversee all company departments within the system. The interface features an "+ Add Department" button for easy creation of new organizational units. Departments are displayed using cards, showing the Department Name, its Active status, and the assigned Department Head, with an action to "View Details" for deeper configuration.

viii. Desktop Settings (User/Admin View)

The Settings page serves as the user's personal hub for customizing application preferences and managing account security. The interface is organized using tabs: Account (for viewing and editing personal details), Security (for managing passwords), and a User Guide. A crucial component of this view is access to Notifications, allowing the user to filter alerts by All, Unread, and Read statuses, along with a Logout action.

ix. Desktop Notifications

This is the central place to track all real-time alerts and updates related to tickets and tasks. The screen displays a clear list of events, such as a comment on a task. Tabs are provided to quickly filter notifications by All, Unread, or Read status.

1.3 Significance of the Study

The system entitled “Development of a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal” seeks to address the increasing demand for an efficient, simplified, and well-organized method of managing service requests and task assignments at the Municipal Information Technology Office. Its relevance is highlighted as follows:

Municipal Information Technology Office (MITO) – As the primary beneficiary, MITO experienced a notable improvement in its task management and ticketing processes. The system provides a centralized platform for logging, tracking, and resolving requests, leading to faster response times, reduced manual effort, and minimized risk of overlooked or delayed tasks. By automating these processes, MITO could focus on delivering higher-quality IT services to the community.

Municipal Staff – The staff at the Municipality of Montalban benefit from faster and more efficient request handling and a more organized and structured workflow. By using the ticketing system, users could conveniently submit their IT-related issues and track the status of their requests, fostering transparency and accountability. Through the task management system, they could easily monitor their assigned tasks, prioritize urgent issues, and manage their workload more effectively. The system’s real-time updates and notifications also help in maintaining clear communication and ensuring that no task goes unattended.

Researchers and Future Developers – This study serve as a valuable reference for future research and development efforts in the field of IT support systems. The methodologies, findings, and challenges encountered during the system’s development could guide students, developers, and researchers in creating similar or more advanced systems tailored to other institutions or organizations.

Educational Institutions – As an academic project, the development and implementation of this system contribute to the practical knowledge and technical skills of the students involved. It provides hands-on experience in software development,

project management, and real-world problem-solving, preparing them for their future careers in the IT industry.

1.4 Scope and Delimitation

This study focused on the development and implementation of a cross-platform Ticketing and Task Management System for the Municipal Information Technology Office (MITO). The system was designed to improve efficiency in handling service requests and provide real-time task monitoring. Key functionalities included user login, ticket management, task management, user management, department management, progress tracking, analytics viewing, report generation, and real-time notifications. The system was accessible via desktop, web, and mobile applications to ensure wider accessibility and convenience for end-users. It was intended for use by MITO personnel, municipal staff, and department heads.

The study was confined to the internal operations of MITO and focused solely on the office's ticketing and task management needs. Non-technical support functions and operations outside MITO were not included. System evaluation was limited to internal users, excluding external clients or third-party service providers. The evaluation of the system centered on its technical performance and usability, based on predefined ISO/IEC 25010 quality criteria. Quantitative data were collected through a structured survey administered during the system deployment phase, with respondents providing numerical ratings using a standardized evaluation instrument. Only these numerical data were included in the analysis, ensuring that the study maintained a purely quantitative approach.

1.5 Operational Framework

The project developers adopted the Spiral Model as the development framework for the MITO Ticketing and Task Management System. Originally introduced by Barry Boehm, the Spiral Model is a risk-driven and iterative software development approach that integrates the structured aspects of the Waterfall Model and the flexibility of the Prototyping Model. Unlike linear approaches, it emphasizes continuous refinement through multiple iterations or “spirals,” where each loop represents a phase of system development. The number of loops is not fixed and may vary depending on the project’s complexity and progress. Each phase of the spiral model passes through all four quadrants: determining objectives, identifying and resolving risks, development and testing, and planning the next iteration. This makes it particularly effective for complex projects where requirements may evolve. (*What Is Spiral Model in Software Engineering?* - *GeeksforGeeks*, n.d.).

The Spiral Model is highly relevant to the development of the MITO Ticketing and Task Management System because the project requires flexibility, risk management, and continuous user feedback. Since MITO’s requirements may change during development, the iterative nature of the model allows for refinement at every stage. Risk assessment ensures that critical concerns such as data privacy, system security, and usability are addressed early. Moreover, constant user involvement ensures that the system remains aligned with MITO’s operational needs, resulting in a reliable and effective solution.

In the **Determine objectives** phase, the project developers collaboratively identify the core requirements and features needed for the system. This includes defining essential functionalities such as ticket creation, task assignment, progress tracking, and real-time notifications. The project developers also define the project scope, timeline, and potential risks. A product backlog is created, listing all features and tasks, which are then prioritized to ensure the most critical and high-impact elements are addressed first.

During the **Identify and resolve risks** phase, the project developers assess potential risks associated with the project, such as system security, scalability issues, and user adoption challenges. Each risk is evaluated for its impact and likelihood, and mitigation strategies are formulated to address them. This phase ensures that any potential roadblocks are anticipated and addressed early in the development cycle.

In the **Development and test** phase, the project developers design and build the system incrementally, implementing features defined in earlier phases. Prototypes and wireframes are created to visualize the system's user interface and user experience, ensuring intuitive navigation and ease of use for municipal staff and requesters. Essential modules like ticket logging, task management, and progress tracking are developed in iterations, with each cycle delivering a functional component of the system. Each component undergoes testing to ensure performance, reliability, and alignment with user requirements. Feedback gathered from MITO staff and test users is then used to refine and improve the system. These insights guide the planning of the next iteration to ensure continuous improvement.

Finally, in the **Plan the next iteration** phase, the project developers evaluate results from testing and user feedback to identify areas for improvement. A working prototype is deployed for real-world testing within MITO, allowing developers to gather practical feedback from end-users. Insights from actual usage are used to refine system functionalities, enhance user experience, and plan subsequent development cycles. Each iteration results in a more stable, user-centered, and efficient version of the system.

1.6 Conceptual Framework

The conceptual framework illustrates the overall process and flow of activities involved in developing the Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office (MITO). It follows the Input-Process-Output (IPO) model, which serves as the structural basis for the study.

The **Input** phase consists of the identified problems of the Municipal Information Technology Office, particularly its reliance on manual methods such as written logs, emails, and verbal communication, which cause delays, miscommunication, and a lack of task visibility. Other inputs consist of the system requirements gathered through consultations, and the essential development tools and system specifications for the system are identified. These technologies are categorized into languages, libraries/frameworks, databases, and development methodologies. The system is developed using Visual Studio Code (VS Code), a lightweight yet powerful integrated development environment (IDE). VS Code was chosen because of its extensive extension ecosystem, debugging capabilities, and Git integration, making it ideal for collaborative and efficient software development. At its foundation, the system leverages HTML5, CSS3, and JavaScript (ES6+). HTML5 structures the content of the application, CSS3 defines the styling and layout with responsive features, and JavaScript enables interactivity and dynamic behavior. Together, these technologies ensure standards compliance, browser compatibility, and broad accessibility before integrating advanced frameworks. Supporting these languages are efficient library and frameworks.

The web and mobile applications are both developed using React (v18.2.0), chosen for its component-based architecture and virtual DOM, which improve performance and scalability compared to traditional JavaScript frameworks. The desktop version is developed using Electron (v28.0.0), which enables cross-platform deployment across Windows, macOS, and Linux. Electron was chosen over native desktop development to ensure code reusability and faster development by leveraging web technologies (React + Material-UI). For data management, MySQL (v3.14.1)

serves as the relational database, chosen for its structured schema, which is critical for maintaining consistency in ticket and task records. Compared to MongoDB, MySQL was preferred because MITO's workflows rely on structured, relational data.

The Spiral Model is adopted as the software development methodology, focusing on iterative development, risk management, and continuous feedback to refine and enhance the system at each phase. By combining these technologies, the system achieves cross-platform functionality, supporting web, mobile, and desktop environments. The selected stack ensures real-time responsiveness, secure authentication, efficient task handling, and an intuitive user interface addressing the requirements of MITO for enhancing ticketing and task management.

These inputs are transformed through the **Process** phase, which involves the development of the proposed system using the Spiral Model of the Software Development Life Cycle. It begins with determining the objectives, where the system's features and functionalities are identified based on the needs of the Municipal Information Technology Office (MITO). Key requirements include a reliable ticketing system for IT service request management, efficient task assignment and monitoring, multi-platform accessibility through web, mobile, and desktop applications, structured data storage using MySQL, and user roles with specific permissions. Once the requirements are clearly defined, the project moves into identifying and resolving risks phase which focuses on analyzing potential challenges such as data security, system performance, and user adoption. Insights from this phase guide the development and testing phase, where system design and implementation take place. This stage involves creating the system architecture, including system flowcharts, an Entity-Relationship Diagram (ERD), and a Data Flow Diagram (DFD) to visualize processes and relationships. User Interface (UI) and User Experience (UX) designs are also developed for web, mobile, and desktop platforms to ensure consistency and ease of use.

During the development stage, the system is built using React for the web and mobile application, and Electron for desktop, all integrated with a MySQL database for centralized and efficient data handling. Afterward, testing is conducted to ensure that performance, security, and usability standards are met. Once validated, the system

proceeds to deployment and implementation, followed by user training sessions to familiarize MITO staff with the system's functionalities. The final quadrant, planning the next iteration, involves gathering feedback and reviewing system performance for further enhancement. Through this iterative and risk-managed approach, the Spiral Model ensures the development of a reliable, secure, and efficient cross-platform ticketing and task management system that strengthens MITO's IT service operations.

Upon the successful completion of the process phase, the **Output** phase results in a fully functional Cross-Platform Ticketing and Task Management System for MITO. It is a comprehensive platform, flexible, accessible, and scalable platform designed to simplify IT support services. The system enables efficient creation, tracking, and management of IT service requests and tasks, ensuring faster response times and improved service delivery. The platform consists of three main applications: a web application for department heads to monitor and assign requests, a mobile application for IT staff to manage tasks in the field, and a desktop application for the super admin with full control and system monitoring capabilities.

Following the output phase, a **feedback loop** is implemented wherein project developers gather insights from municipal staff and other stakeholders to refine the system and address any issues that may arise post-deployment. The system's effectiveness is evaluated based on the ISO/IEC 25010 Software Quality Model, focusing on functional suitability, compatibility, interaction capability, reliability, and security. This iterative process ensures that the system remains reliable, user-centered, and aligned with the operational needs of the Municipal Information Technology Office.

1.7 Definition of Terms

Cross-Platform – The ability of the system to operate on multiple environments (web, mobile, and desktop) while maintaining data synchronization and consistent functionality.

ISO/IEC 25010 – An international standard for software quality evaluation used in this study to assess the developed system's performance in terms of functionality, compatibility, reliability, security, and interaction capability.

MITO (Municipal Information Technology Office) – The government office of Montalban, Rizal responsible for managing the municipality's information technology infrastructure, support services, and system development.

Task Management System – A software component that allows users to create, assign, prioritize, and monitor tasks to ensure timely and organized workflow execution.

Ticket – A recorded service request or issue submitted by a user requiring attention, resolution, or action from the IT department.

Ticketing System – A software tool used to log, track, and manage service requests from creation to resolution, improving organization and response time.

CHAPTER II

REVIEW OF RELATED LITERATURES, STUDIES, AND SYSTEMS

In this chapter, various local and foreign literatures, studies, and systems were reviewed by the researcher to gain an understanding of existing research and other academic works relevant to the area of study, and thus, provide an in-depth understanding of Ticketing and Task Management Systems.

2.1 Web Application Development: Ticketing Management System

Gumarang et al. (2023) on the "ICT Helpdesk Ticketing Management System" serves as a vital architectural precedent for the proposed MITO: Ticketing and Task Management System. Their study aimed to implement a unified online platform to simplify IT service requests at St. Paul University Philippines (SPUP), effectively eliminating the inefficiencies of manual processes, such as reliance on phone calls and physical tracking, which resulted in pending issues and poor record-keeping. The system's successful architecture, defined by two distinct user panels, the Customer (for problem reporting) and the ICT Technical Staff (for management, tracking, and report generation), validates the necessity of a role-based access control system fundamental to effective task and management platforms. The functions of the ICT Technical Staff panel, including comprehensive ticket oversight and systematic tracking of resolution status, directly parallel the multi-tier administrative and workflow functionality required by the MITO system. The successful implementation of this centralized, user-centric, and efficient web-based solution highlights the importance of modular design and process digitization for enhancing service accessibility and efficiency within a specialized domain. However, a significant gap exists in the application domain and core system purpose: Gumarang et al. (2023) focused on an internal institutional system for managing IT-related problems, where the "ticket" is purely an internal technical support request. Conversely, the MITO: Ticketing and Task Management System is an administrative and service platform for a community partner municipal office, where the "ticket" represents a public service request or complaint. This critical difference shifts the focus from internal technical support to public-facing governmental service and administrative workflow, requiring MITO to integrate essential features for

external public complaint submission, inter-departmental task delegation, regulatory compliance, and public transparency, functionalities not relevant to an internal university helpdesk system.

Wonohardjo et al. (2022) reported that PT DENSO Indonesia faced significant challenges with its legacy helpdesk ticketing system, which was based on IBM Lotus Notes. This system was deemed inadequate due to its limited features, strong dependence on the platform provider, and the difficulty in modifying its source code. As a result of the company's decision to terminate its Lotus Notes subscription, a new, independent solution became necessary. To address these issues, the researchers undertook a project to design and implement a new web-based helpdesk ticketing system. This new system was developed using the Extreme Programming (XP) software development methodology for its flexibility and rapid iteration, and Unified Modeling Language (UML) for detailed design documentation. The primary goal was to create a system that was not only functional and reliable but also easily maintainable and scalable, offering a long-term solution for PT DENSO Indonesia. To evaluate the new system's effectiveness and efficiency, the researchers utilized quantitative metrics, specifically calculating the completion rate and time-based efficiency. A problem reporting scenario with 15 participants was used to test the system's performance. The results showed successful implementation, with the web-based system achieving an 86% completion rate and a time-based efficiency of 0.00924 goals/sec. These findings confirmed that the new system effectively resolved the shortcomings of the previous Lotus Notes system, providing PT DENSO Indonesia with an independent, easily modifiable platform for managing help desk tickets. This study directly supports the development of the "MITO: Ticketing and Task Management System" by providing a successful case study of how to replace an outdated, platform-dependent system with a new, independent, and efficient web-based solution. The researchers' use of XP and UML for development and their quantitative validation of the system's success offer a valuable blueprint and a strong precedent for the MITO project's objectives of creating a functional, reliable, and easily maintainable platform.

According to Mission (2021), "multi-channel support and ticketing interface for online support management system platforms," focused on developing and evaluating

a system to simplify support requests from various communication channels into a single, cohesive interface for university operations. Anchored in research and development, the project successfully created a system that allows technical support agents to efficiently track, prioritize, and resolve queries, with a subsequent evaluation using the ISO/IEC 25010 software product quality models confirming high stakeholder satisfaction across functional suitability, usability, and maintainability. This previous research provides a validated model for centralized, multi-channel ticket management and demonstrates the effectiveness of such a system in a large organizational setting, specifically a university. This foundational success in centralizing support is directly relevant to the current "MITO: Ticketing and Task Management System," as both projects share the core objective of improving organizational efficiency through digital ticket handling. However, a significant gap exists because the previous study implemented a unified, primarily web-based system for general institutional support, whereas the current research, MITO, addresses a distinct need for role-specific, multi-platform deployment (Web for Department Heads, Mobile for Municipal Staff, and Desktop for the Administrator) tailored for a municipal partner, requiring a greater complexity in integrated task assignment, employee management, and real-time progress tracking across devices, which the predecessor system did not fully detail or evaluate.

According to Prastama et al. (2023), the research aimed to analyze the user experience of the Open-Source Ticket Request System (OTRS) utilized by PT Datacomm Diangraha, an Indonesian IT service provider. The study specifically examined the Dashboard and Create Ticket Menu, which were perceived by users as lacking efficiency. By employing a heuristic evaluation methodology, the researchers evaluated the interface design using 10 heuristic principles. Data collection involved a Google Form questionnaire distributed to 60 employees from the Service Help Desk and Service Management Division, resulting in 55 responses. The analysis revealed nine areas requiring improvement, and the researchers subsequently proposed User Interface (UI) enhancement recommendations to optimize ticketing and monitoring processes within OTRS. The company acknowledged and positively received these recommendations, indicating potential for significant improvements and innovations within the OTRS system. Prastama et al. provides a critical UX analysis framework for

the MITO: Ticketing and Task Management System. The study demonstrated that even a robust system like OTRS could suffer from perceived inefficiencies due to a suboptimal user interface. By employing heuristic evaluation methodology and quantitative feedback from users, the researchers were able to identify and propose specific UI enhancements that would optimize ticketing and monitoring processes. This research directly underscores the necessity of a user-centered design approach for the MITO project, emphasizing that a system's efficacy is contingent not only on its functionality but also on its usability. The study validates that rigorous UX analysis is essential for identifying actionable improvements, ensuring the final system is intuitive and efficient for its end-users. A key gap, however, lies in the application context of the evaluation. Prastama et al. (2023) focused on analyzing and improving an existing system (OTRS) within a specialized IT service provider, dealing primarily with internal technical support tickets. The project, MITO, is the development of a new system for a community partner municipal office. This difference introduces a gap where the UX framework must be applied from the initial design phase for a broader user base (citizens and municipal staff) and a wider variety of inquiries (public service requests, non-technical complaints, and administrative tasks). While they focused on UI improvements, MITO also integrates a crucial Task Management component, meaning the system's UX analysis must be expanded to ensure the efficient management of both tickets and internal office workflows, a complexity not addressed by their evaluation of a dedicated ticketing system.

According to Jain et al. (2025), the research titled "AI Enhanced Ticket Management System for Optimized Support" proposes a unified, AI-driven microservices architecture to enhance operational efficiency in various ticket management domains, such as IT Service Management (ITSM) and Security Operations (SOC), with the core objectives of meeting SLAs, reducing costs by reducing redundancy, and prioritizing tickets. The proposed system is application-agnostic, integrating efficiently with existing tools like ServiceNow and Jira, offering modular microservices for clustering, prioritization, and resolution recommendation. A key innovation is the integration of spatial and temporal factors (like device topology, timings, and data source) into traditional semantic similarity algorithms to achieve high-fidelity ticket grouping, which resulted in a Rand score of 0.96 in testing,

underscoring the importance of domain knowledge in achieving high fidelity in ticket grouping. The robust methodology includes data ingestion via APIs, feature extraction using the Hugging Face BGE model, and a highly customizable clustering component where administrators could tune parameters like hierarchical weight and semantic similarity thresholds to meet specific organizational needs. Furthermore, the system leverages Generative AI for resolution generation, validated against organizational knowledge bases, with 95% of suggested resolutions approved by engineers. The deployment achieved significant operational improvements, including a 30% reduction in alerts and an enhancement of key performance indicators (KPIs) like Mean Time to Solve (MTTS), affirming the system's ability to provide an adaptable, robust, and highly efficient solution for modern IT support operations. This research provides a cutting-edge conceptual and technical model for the MITO: Ticketing and Task Management System by strongly validating the effectiveness of an AI-driven, microservices-based approach to achieve high operational efficiency, particularly through intelligent clustering, customization dashboards, and automated resolution features, underscoring the necessity of contextual factors (spatial and temporal) in grouping tasks, a principle relevant to managing inter-departmental workflows in the MITO system. However, a significant gap exists in the application domain and technological focus: Jain et al. (2025) are focused on highly technical, enterprise-level ITSM and SOC environments, dealing with IT infrastructure incidents and leveraging advanced LLMs and Generative AI for automation, while the MITO system is a practical development for a community partner municipal office, where the primary challenge is managing human-centric administrative and public service requests (non-technical tickets), meaning the gap is the immense difference between optimizing IT infrastructure incident resolution using bleeding-edge AI and developing a reliable, user-friendly system for public accountability and administrative workflow management in local governance.

According to Hampf (2021), the thesis titled "Development of a Task Management Application - a Progressive Web Application" documents the creation of a cross-platform application designed to manage tasks stored in a specific plaintext format, `todo.txt`, residing on a server. The purpose was to provide a resilient, synchronized task manager usable across different platforms, allowing for interaction

with data using various tools. The solution consists of three main parts: a reusable JavaScript library for parsing the todo.txt format, a Progressive Web Application (PWA) for the graphical user interface (built using JavaScript, HTML, CSS, and the Vue.js framework), and a small Nextcloud application for hosting and interaction. The use of a PWA ensures cross-platform compatibility and enables the application to function reliably, engaging users with a platform-native experience, even offline, by leveraging client-side storage like localForage (a wrapper for IndexedDB). The PWA interacts with the server-side plaintext file using either the Nextcloud API or the WebDAV protocol, ensuring flexibility and independence from a specific host environment. This study provides valuable technical validation for the MITO: Ticketing and Task Management System by confirming the viability and benefits of developing a Progressive Web Application (PWA) for cross-platform task management, which directly informs MITO's design requirement for mobile and desktop usability. Furthermore, the use of Vue.js and the modular approach to handling file parsing and data synchronization provides a strong architectural blueprint for MITO's front-end design. However, a significant gap exists in the application complexity and data structure: Hampf (2021) focuses on a personal task manager built for the simple, plaintext todo.txt format, stored on a private server (Nextcloud). The MITO system is a professional, team-based platform for a municipal office that must handle complex public service requests (ticketing) with relational data, multiple users, specialized roles, and adherence to strict public accountability standards. The gap requires adapting the proven principles of PWA development and cross-platform compatibility from a single-user, plain text-file context to a multi-user, database-driven, highly structured administrative system.

2.2 Desktop Application Development: Ticketing and Task Workflow Systems

Lindblom (2024) thesis examined the critical role of ticketing systems in boosting the efficiency of IT service desks, specifically analyzing how a client company's existing system, ConnectWise Manage, affected its operations. The main goal was to understand this impact and pinpoint strategies to improve the service desk's speed. To achieve this, Lindblom closely examined all key aspects of the service desk's functions, including staff duties, work procedures, ticket handling processes, ticket

types, and user requests, paying special attention to ConnectWise Manage's features to determine what makes ticket management effective things like automation for routine tasks, accurate ticket sorting, and fast queue management. The research found several areas for improvement, concluding that connecting call systems directly with the ticketing platform, implementing smarter automation workflows, and adopting a more user-friendly design were effective ways to increase speed and reduce resolution times. The thesis ultimately stressed that while the ticketing tool is important, maximizing its potential requires a holistic approach that includes smooth connections with other systems, strategic automation, and a strong focus on user experience. These findings offer important guidance for the design of the MITO: Ticketing and Task Management System, highlighting automation, integration, and user-friendliness as key drivers of efficiency. Despite this guidance, a major difference lies in the application context and system scope: Lindblom focused on a private IT service company dealing with technical service tickets on an existing commercial platform, whereas MITO is a new system for a municipal office that must handle public service requests and administrative tasks (non-technical tickets) and adhere to public governance and transparency rules, making the solution much wider and more complex as it integrates a Task Management system to handle all internal office workflows alongside ticketing.

According to Gayathri et al. (2022), this research introduces a desktop application designed to simplify file management in cloud computing environments, prioritizing resource optimization and enhanced security. The application uses virtual machines for efficient resource allocation and scalability, addressing load balancing through dynamic adjustments. It features distinct admin and user modules, where administrators manage file uploads, user details, and approvals, while users could register, view files, and request downloads. To ensure data security, a cryptographic algorithm is implemented, and file allocation is optimized using Grey Wolf and Priority Based Optimization techniques, which prioritize frequently accessed files to reduce waiting times. The system utilizes three cloud storage services to create a virtualized storage pool accessible via the internet, enhancing scalability and flexibility. This approach aims to provide a cost-effective solution for managing files in the cloud, focusing on efficient resource utilization and strong security measures. By optimizing file allocation and prioritizing high-demand files, the application minimizes approval

times and enhances overall system responsiveness, making it a valuable tool for organizations seeking to improve their cloud storage strategies. This study provides a valuable technical precedent for the development of the "MITO: Ticketing and Task Management System" by demonstrating a successful model for utilizing cloud computing for scalable storage and implementing rigorous security measures and priority-based optimization techniques to enhance system responsiveness and efficiency, all of which are critical considerations for a municipal office application. However, a major gap exists in the application focus and system functionality. Gayathri et al. (2022) are exclusively focused on file management and cloud storage optimization, addressing technical issues like load balancing and cryptographic security for digital assets. The system, MITO, is a Ticketing and Task Management System for a municipal office, where the primary function is administrative workflow management handling citizen requests (ticketing) and internal assignments (task management) not file storage optimization. The gaps are therefore in core purpose, which means the principles of security and prioritization must be adapted to the administrative and service-oriented nature of public sector work, which involves tracking service status, managing personnel workflows, and ensuring public transparency, a domain separate from the technical challenges of virtualized cloud storage.

According to K. D. Aglibar & Rodelas (2022) This research investigates the impact of critical and auto-generated tickets on management and worker productivity within a technical support environment, particularly in the Information Technology industry. The study highlights that ticketing systems, like Service Now, are crucial for connecting companies with clients and managing support operations. Critical tickets, which require immediate resolution under strict service level agreements (SLAs), place significant pressure on support resources, affecting their ability to manage other tasks and impacting overall productivity. Auto-generated tickets, while often routine, could also contribute to stress and workload challenges, especially when they escalate into critical issues. The researchers used data collected from management, support workers, and the Service Now ticketing system to analyze these impacts. They found that knowledge management is essential for helping support staff quickly resolve incidents and develop permanent solutions. The study concludes that critical and auto tickets significantly affect daily productivity, particularly when onboarding new employees

during ongoing critical incidents. To mitigate these challenges, the authors recommend implementing efficient knowledge management systems and dashboards to monitor SLAs and incoming tickets, ensuring better documentation and tracking of solutions. They also suggest further research into the mental health impact of these tickets on support staff and their direct impact on business operations, as well as exploring how knowledge management could be effectively utilized to navigate complex troubleshooting scenarios. The research by K. D. Aglibar & Rodelas provides a crucial analysis of the impact of critical and auto-generated tickets on worker productivity, offering a clear set of functional requirements for the "MITO: Ticketing and Task Management System." The study underscores that managing strict Service Level Agreements (SLAs) and high-volume ticket flows could negatively affect support resources. The authors' key recommendations, the implementation of efficient knowledge management systems and real-time dashboards for monitoring incoming tickets and SLAs, are directly applicable to the MITO project. This research validates the necessity for the MITO system to be designed not only for ticket processing but also for mitigating productivity challenges by providing staff with integrated resources and management with the tools for real-time operational oversight. The primary gap between this study and the work, however, is the fundamental application environment, and this is a formal one, K. D. Aglibar & Rodelas (2022) focus on an IT service provider dealing with technical support issues and auto-generated system alerts, using a commercial system (Service Now), whereas the MITO system is a new platform developed for a municipal office, where "critical tickets" relate to urgent public service requests, complaints, and non-technical administrative issues, not IT infrastructure problems. This shift in context requires the system to address the unique challenges of public sector accountability, citizen communication, and adherence to government-specific internal workflows, which includes the integration of a Task Management component alongside ticketing, a functional requirement that extends beyond the purely technical help desk scope of the existing literature.

According to Huang et al. (2024), this research focuses on enhancing power grid management through a smart two-ticket system that utilizes edge computing. The study addresses the challenges posed by the increasing integration of Internet of Things (IoT) technology into power industry control systems, which results in a vast

deployment of intelligent terminals and sensing devices within smart grids. This influx of devices leads to a "dual heterogeneity" and "virtual reality" environment, where traditional communication systems struggle with reliability and disaster response, often relying heavily on base stations that could fail, causing local network interruptions. To overcome these issues, Huang et al. propose an intelligent IoT system based on edge computing, which is designed to handle dynamic control, strategy management, and data preprocessing at the network's edge. This approach aims to reduce network transmission capacity, lower operating costs, and optimize equipment maintenance. The research highlights the importance of edge computing in achieving information acquisition, operational control, analysis, and decision support through target perception and control, combined with wireless communication. Simulation experiments were conducted to evaluate the system's performance, demonstrating that edge computing significantly reduces data unloading completion times compared to cloud-based and random allocation methods. Specifically, when the network included 70 Centralized Units of Execution and 100 Heterogeneous Units of Execution, the proposed schemes "HUE Cloud Edge" and "HUE-Cloud Edge GM" achieved unloading completion times between 0.25-0.45 and 0.2-0.4, respectively, showcasing the efficiency and scalability of the edge computing approach in power grid management. This study, while technologically advanced, offers a high-level conceptual validation for the "MITO: Ticketing and Task Management System" by successfully designing a "two-ticket system" to manage complex, time-sensitive operational processes, emphasizing efficiency, reliability, and decentralized control, which are core goals for the system. However, a crucial gap exists in the application domain and technological focus. Huang et al. (2024) are centered on the power industry's smart grid, where the "tickets" manage physical IoT devices, data unloading, and real-time control using highly specialized technologies like edge computing for computational speed. The MITO system, in contrast, is an administrative platform for a community partner municipal office, where the "tickets" manage citizen service requests, complaints, and internal office tasks, not power grid assets. The gaps are therefore the profound differences in context and purpose which means translating the principles of efficiency and system reliability demonstrated in their specialized industrial/IoT environment to the human-centric, public service domain, where the challenges involve managing diverse administrative workflows, ensuring staff

productivity, and maintaining public accountability, rather than optimizing data transmission and control unit execution.

According to Gupta et al. (2022), a study was undertaken to address the computationally challenging problem of task scheduling optimization within cloud computing environments, specifically focusing on minimizing the make span, or total completion time. Their research centered on refining the Heterogeneous Earliest Finish Time (HEFT) algorithm, a widely recognized method for scheduling tasks in diverse computing environments. Recognizing that the basic HEFT algorithm, which relies on average computation costs and the selection of the first available idle slot, may not consistently yield optimal results, they proposed and evaluated modified versions of the algorithm. Through rigorous experimentation, they explored the impact of different rank generation and processor selection schemes, including the use of maximum and minimum computation costs for task prioritization, and the selection of the idle slot that minimizes finish time. Their findings revealed that these modifications significantly influence the makespan, demonstrating that both rank calculation and processor selection strategies play critical roles in scheduling efficiency. Ultimately, Gupta et al. (2022) concluded that enhanced versions of the HEFT algorithm, particularly those incorporating the selection of the best idle slot, exhibit improved performance compared to the basic HEFT algorithm, offering valuable insights for optimizing task scheduling in cloud environments. This research provides a theoretical and algorithmic foundation for the "MITO: Ticketing and Task Management System" by underscoring the critical need for sophisticated task scheduling and optimization to achieve efficiency and minimize completion time, which directly aligns with MITO's goal of optimizing staff workflows. The study validates that intelligent prioritization and resource allocation (processor selection) are essential to efficiency. However, a significant gap exists in the tasks and resources, and this is a formal one. Gupta et al. (2022) focus exclusively on computational task scheduling (software processes) in a cloud computing environment, optimizing for metrics like makespan and computation cost. The system, MITO, is designed for a municipal office where the "tasks" are human-centric administrative duties and service requests, and the "resources" are municipal staff, not virtual processors. The gaps require the translation of their principles of algorithmic optimization to the complexities of a human workflow environment, where

scheduling must account for factors like staff specialization, real-world task dependencies, and the subjective urgency of public inquiries (tickets), rather than purely computational metrics.

Montecillo et al. (2024) investigated the potential of a Task Management and Productivity App integrated with a Desktop Robot Display ("Pixie") to help university students manage their academic workload and enhance productivity. Recognizing the challenges students face with task management, social isolation, and mental well-being, the researchers developed a system that combines a mobile application for task management (checklists, reminders, notes, and a game) with a physical robot that visually displays this information. Drawing on the Technology Acceptance Model (TAM), the study evaluated the acceptance level and perceived effectiveness of this system among 50 computer science students. The findings indicated a positive reception, with students perceiving the system as easy to use and beneficial for improving their productivity and overall academic performance. This research contributes to the understanding of how companion-like robots integrated with productivity apps could be accepted and potentially beneficial tools for students in managing their tasks and mitigating issues like social isolation. This study provides valuable conceptual support for the "MITO: Ticketing and Task Management System" by demonstrating the positive user acceptance and perceived effectiveness of a combined Task Management application that is designed to improve productivity and manage workload, validating the core functionality of the system. The use of the Technology Acceptance Model (TAM) also offers a clear framework for evaluating the user acceptance of the MITO platform. However, a significant gap exists in both the application context and technological complexity. Montecillo et al. (2024) Focus on the highly specific and niche environment of university students and their academic workload, using a highly specialized robot-integrated system to address student-specific issues like social isolation. In sharp contrast, the project, MITO, is a platform for a community partner municipal office, where the system must address the complexities of public service administration, inter-departmental workflows, and public accountability, and this is a formal one, not academic deadlines or student well-being. The gaps are the profound differences in system purpose and technological scope, which means adapting the principles of task management and productivity to a

formal public-sector setting without the use of specialized hardware like a robot display, focusing instead on administrative efficiency, public service request (ticketing) management, and robust internal task delegation.

According to Al-Hawari & Barham (2021) The research titled "A machine learning based help desk system for IT service management" introduces a help desk system designed to act as a single point of contact (SPOC) between users and IT staff, with a primary goal of minimizing ticket resolution time, saving resources, and enhancing user satisfaction. The system utilizes an in-house developed, accurate machine learning classification model to automatically associate incoming tickets with the correct IT service from the start, a feature identified as critical for minimizing misclassification and subsequent delays. The model was generated through an empirical methodology including ticket data preprocessing, word stemming, feature vectorization, and machine learning algorithm tuning. The authors found that including the ticket comments and full description in the training data was a key factor that boosted the model's prediction accuracy significantly, from 53.8% to 81.4%. Functionally, the system supports both an Admin view for defining services, managing user roles, and generating management reports (including measuring ITIL KPIs), and a user view for reporting issues, requesting services, and exchanging information (including files and comments) with IT staff. The study provides a crucial validation for the MITO: Ticketing and Task Management System by demonstrating the successful development and positive impact of a system that integrates automatic classification and KPI measurement within an administrative context, directly informing MITO's design goal of optimizing staff workflows and improving service delivery. However, a significant gap exists in the application domain and core functionality: Al-Hawari & Barham (2021) Focus exclusively on an IT Service Management (ITSM) system for a university, dealing with technical IT incidents and relying on machine learning to categorize technical issues. The MITO system is a broader Ticketing and Task Management System for a community partner municipal office, where the tickets are predominantly non-technical public service requests and administrative duties. This shift requires the adaptation of the proven principles of automatic classification and workflow management from the specialized, technical ITSM context to the unique complexities of public accountability, citizen

communication, and non-technical administrative task delegation in local government.

K. D. M. Aglibar et al. (2023) investigated the effectiveness of ticketing systems, particularly JIRA, for managing projects and tracking issues in IT companies. Recognizing the high volume of daily transactions and the need for meticulous record-keeping, the study aimed to determine JIRA's utility for both support teams and internal company members. Using a descriptive research method, the authors analyzed existing case studies and journals to evaluate JIRA's advantages, disadvantages, and operational mechanisms. The research concluded that the JIRA Ticketing System is beneficial for both support groups and internal stakeholders by providing a robust platform for tracking incidents and requests, despite its limitations. The authors recommended further research using other analytical models to uncover new insights for both established and startup IT companies in selecting efficient ticketing systems. The study highlighted key features of an effective ticketing system, such as collaboration capabilities, multi-channel support, convenience, quick issue resolution, and detailed reporting and analysis. It also touched upon metrics for measuring effectiveness, including ticket volume and resolution time, and underscored the inefficiencies of manual tracking and reporting. The review of related literature provided an overview of JIRA's evolution from a bug tracking system to a versatile tool, noting its customizability and successful implementation in companies like NIF ICCS. While acknowledging JIRA's limitations, the research ultimately affirmed its helpfulness in managing tickets and supporting project deliverables. This descriptive research provides a strong conceptual and functional blueprint for the "MITO: Ticketing and Task Management System" by validating the core features essential for an effective platform, specifically collaboration, reporting, and efficient issue resolution, and affirming the utility of a consolidated system over manual tracking. However, a significant gap lies in the application context and system scope, and this is a formal one. K. D. M. Aglibar et al. (2023) The study is focused entirely on the private, commercial IT industry and an existing tool (JIRA) for bug tracking and technical support, whereas the MITO project is the ground-up development of a new system for a community partner municipal office. This shift introduces the unique gaps of public sector requirements, demanding that the system manages citizen service requests, public transparency, and administrative tasks that are non-technical in nature, and

integrates a comprehensive Task Management component for all office duties, a functional breadth not covered by the IT-centric issue-tracking focus of the existing literature.

2.3 Mobile Application Development: Task Management and Automation

Ahmetoglu et al. (2024), investigated the disconnect between psychological research on time management and the design of personal task management applications. Recognizing that individuals frequently underestimate task duration due to the "planning fallacy," a cognitive bias, the authors aimed to identify strategies to improve time estimation accuracy within these apps. Through a two-pronged approach, they first reviewed psychological literature to pinpoint effective time estimation techniques and then conducted a functionality analysis of popular task management apps to assess their implementation of these strategies. Their findings revealed a significant discrepancy, demonstrating that while research-informed strategies exist, they are seldom effectively integrated into current app designs. This gap highlights a crucial area for improvement, suggesting that future development efforts should prioritize aiding users in accurate task duration estimation. The study underscores the need for a more research-driven approach to task management app design, with the potential to enhance user productivity and reduce work-related stress. This study offers a vital research-driven design principle for the "MITO: Ticketing and Task Management System" by affirming that effective Task Management goes beyond simple checklists and must integrate strategies to mitigate cognitive biases like the planning fallacy, directly informing the goal of enhancing staff productivity through accurate task duration and deadline management. However, a significant gap exists in the application context and scale of impact, and this is a formal one. Ahmetoglu et al. (2024) Focus on personal task management apps and the individual's planning fallacy, whereas the MITO system is being developed for a community partner municipal office, a highly professional and collaborative environment. This shift introduces the gap of needing to adapt these psychological principles for a team-based system where task duration and resource allocation must be accurately estimated and managed across multiple staff members and administrative workflows (Task Management) that are integrated with public-facing service requests (Ticketing), a complexity that extends beyond individual

planning and into organizational efficiency and public accountability.

Lee et al. (2024) introduced an innovative approach to mobile task automation using Large Language Models (LLMs). Recognizing the potential of LLMs to automate complex and repetitive mobile tasks, they also addressed the limitations of unreliability and high operational costs. To overcome these challenges, they developed MobileGPT, an LLM-based system that emulates human cognitive processes when interacting with mobile apps. This system operates through four key stages: explore, select, derive, and recall, effectively breaking down tasks into modular sub-tasks that could be reused and adapted. By integrating a "human-like app memory," MobileGPT enhances the precision and efficiency of task learning. Through evaluations on a dataset of 185 tasks across 18 mobile apps, they demonstrated that MobileGPT achieves 82.7% accuracy in automating and learning new tasks, and 98.75% accuracy in adapting tasks to different contexts. Moreover, MobileGPT significantly reduces latency and cost compared to a GPT-4 powered baseline, marking a substantial advancement in practical mobile task automation. This research provides an advanced, forward-looking conceptual framework for the "MITO: Ticketing and Task Management System" by demonstrating the potential for sophisticated task automation, modular task learning, and high efficiency through a specialized system design, which informs the long-term potential for enhancing the municipal office's operational speed. However, a significant gap exists in the application and core technology, and this is a formal one. Lee et al. (2024) work is focused on Mobile Task Automation for mobile applications using advanced Large Language Models and edge computing principles, optimizing for latency and cost in a highly technical domain. The project, MITO, is the development of a web-based administrative system for a community partner municipal office, where the primary challenge is managing human workflows for public service requests (ticketing) and internal administrative duties (task management), and not the automation of mobile app interactions with an LLM. The gaps are therefore the immense difference in technological complexity and purpose, which means solving organizational and administrative inefficiencies using conventional web technologies, while their study addresses cutting-edge problems in AI-driven mobile automation, highlighting a chasm between the practical needs of local governance and the current frontiers of academic research.

Zhao et al. (2023) investigated the effectiveness of productivity tools in mitigating academic procrastination among students. Recognizing procrastination as a significant impediment to academic performance and mental well-being, they developed an academic planning and time management application, Proccoli, to explore how such tools influence student behaviors. Their study aimed to bridge the gap in understanding how these applications aid students in overcoming procrastination, particularly by examining which app features benefit specific student profiles. By analyzing student perceptions of Proccoli, modeling procrastination behaviors through Hawkes process mining, and assessing in-app activities using planning and performance metrics, the authors uncovered valuable insights. Their findings highlighted the necessity for personalized self-regulation support within Proccoli, as students with varying in-app studying behaviors exhibited diverse perceptions of the app's functionalities. Furthermore, they found a significant correlation between social accountability prompts provided by Proccoli and reductions in student procrastination. Ultimately, this research emphasizes the importance of tailoring productivity tools to individual students to effectively combat academic procrastination. This study offers crucial user-centric design principles for the "MITO: Ticketing and Task Management System" by demonstrating the necessity of personalized support and social accountability features to enhance user adoption and address human factors like procrastination, which directly impacts staff productivity in the municipal office. However, a significant gap exists in the application context and scale of operation, and this is a formal one. Zhao et al. (2023) focused on a personal planning app for university students to manage academic tasks, where the motivation and metrics are individual academic performance. The project, MITO, is a professional, team-based administrative system for a community partner municipal office, where the system must address organizational inefficiency in managing public service requests (ticketing) and official tasks, not individual student procrastination. The gaps are therefore the shift from individual academic self-regulation to collective administrative efficiency and public accountability, requiring the adaptation of the principles of personalization and accountability to a formal, team-based government setting.

Song et al. (2024) presented a novel framework for mobile task automation that

leverages both vision-based user interface (UI) understanding and Large Language Model (LLM) task planning. Addressing the limitations of traditional methods like Programming By Demonstration (PBD), which struggle with app updates and predefined tasks, and the shortcomings of view hierarchy-based approaches, which face accessibility and structural issues, the authors introduced VisionTasker. This two-stage system first converts UI screenshots into natural language interpretations, eliminating the need for view hierarchies. Subsequently, it employs a step-by-step task planning method, presenting interfaces sequentially to the LLM to identify relevant elements and determine the next action. Through extensive experiments, VisionTasker demonstrated superior performance compared to previous methods, effectively representing UIs across diverse datasets. Moreover, when automating real-world tasks on Android devices, it outperformed humans in unfamiliar scenarios and exhibited significant improvements when integrated with PBD. By open-sourcing VisionTasker, the authors provide a valuable tool for advancing mobile task automation. This research provides a cutting-edge conceptual model for advanced task planning and workflow interpretation that informs the long-term potential of the "MITO: Ticketing and Task Management System" by validating the power of sequential, step-by-step task decomposition and planning, which is a crucial process for managing complex municipal workflows. However, a major gap exists in the application and core technology, and this is a formal one. Song et al. (2024) are focused on AI-driven mobile task automation for consumer applications, relying on highly specialized technologies like Large Language Models and vision-based UI analysis. The project, MITO, is a web-based system development for a community partner municipal office, where the primary challenge is managing human-centric administrative workflows for public service requests (ticketing) and internal tasks. The gaps are the profound difference in technological approach and system purpose, which means solving organizational and administrative inefficiencies using conventional, accessible web platforms, while their study addresses the frontier of AI and mobile device interaction, highlighting the need for research to focus on practical, easily deployable solutions for local governance rather than bleeding-edge AI for automation.

Martín-García & Pérez Marín (2020) explored the potential of gamification in task management for university students. Recognizing that students often struggle with

time management and procrastination, they hypothesized that a gamified mobile application could help them view tasks as challenges and utilize time more effectively. To test this, they developed a prototype of the "Fun Task Manager" app, designed to integrate gamified elements into daily task management. The app aimed to remind students of their tasks and incentivize completion through rewards, effectively transforming task completion into a game-like experience. The authors detailed the design process of the prototype and evaluated it with 90 first-year Primary Education university students, who validated the app's concept and functionality. Their findings suggest that a gamified approach to task management could be a viable strategy to combat procrastination and improve time utilization among university students. This study offers a valuable design strategy for the "MITO: Ticketing and Task Management System" by demonstrating the positive impact and validation of gamification as a technique to enhance task completion, motivation, and time management through a mobile application, which could inform optional incentive features for municipal staff productivity. However, a significant gap exists in the application context and motivational scope, and this is a formal one. Martín-García & Pérez Marín (2020) focused on an individual, academic setting for university students, where the goal is to combat procrastination with a "Fun Task Manager." The project, MITO, is a professional administrative system for a community partner municipal office, where the motivation is driven by public service, professional duty, and public accountability, not academic performance or game-like rewards. The gaps require the adaptation of the general principles of motivating task completion from a student-centric, personal gamified context to a professional, public-sector, team-based environment that manages both citizen ticketing and official tasks, focusing less on personal fun and more on professional efficiency and compliance.

Zhang et al. (2024) introduced a new evaluation framework designed to address the challenges of assessing mobile UI task automation agents. Recognizing the limitations of existing evaluation methods, which often rely on human validation or predefined action sequences, the authors developed LlamaTouch to provide a more scalable and faithful testbed. The core innovation of LlamaTouch is its focus on evaluating whether an agent could correctly traverse essential application and system states during task execution. By employing on-device task execution, fine-grained UI

component annotation, and a multi-level application state matching algorithm, LlamaTouch accurately assesses agent performance in real-world mobile environments. The testbed includes four mobile agents and 496 diverse tasks, demonstrating its ability to evaluate agents effectively across various applications. Evaluation results confirmed LlamaTouch's high faithfulness and superior scalability compared to human validation. Furthermore, the testbed facilitates easy task annotation and the integration of new mobile agents, making it a valuable resource for advancing mobile UI task automation research. This study provides an advanced conceptual validation for rigorous, state-based system evaluation for the "MITO: Ticketing and Task Management System," emphasizing the necessity of accurately verifying the correct traversal of system states during complex task execution, which is a key requirement for managing municipal workflows. The research method of ensuring faithfulness and scalability in evaluation is a principle directly applicable to testing the reliability of MITO's ticketing and task processes. However, a significant gap exists in the application and core technology, and this is a formal one. Zhang et al. (2024) Focus on the highly specialized, cutting-edge domain of mobile UI task automation agents, leveraging deep technical components like fine-grained UI annotation and application state matching to test AI agents. The project, MITO, is the development of a web-based administrative system for a community partner municipal office, where the primary challenge is managing human-centric administrative workflows for public service requests (ticketing) and internal tasks. The gap is the vast difference between evaluating AI-driven mobile automation and developing a user-friendly, functionally reliable system for local governance, which means adapting their principles of rigorous state-based evaluation to the more conventional, yet equally critical, environment of administrative web applications and public accountability.

Al-Qassab (2021) focused on analyzing and digitalizing employee-generated data collected from their mobile devices. The project aimed to create a system integrated into Azure DevOps and Common Data Service environments to initiate reclamations and minimize errors. The resulting system was designed for usability on both mobile and desktop platforms, with core functionalities including data gathering and handling of information such as start time, end time, duration, and data holder. This data was stored in the commissioner's Common Data Service cloud for subsequent

analysis. The implementation of this system for A-A White Blue Consulting OY Company facilitated efficient work management and improved accuracy in project organization, aiding in the distribution of tasks among employees. While the system was tested in a typical environment for a limited duration, its capacity to manage high-pressure workloads was not assessed. Future development suggestions included comprehensive testing and potential expansion of the system to connect with Jira and Azure DevOps systems. This study provides a valuable case for the "MITO: Ticketing and Task Management System" by demonstrating the successful digitalization of worktime and task management for efficient task distribution and project organization among employees, confirming the viability of a system focused on tracking task duration and project accuracy. Furthermore, the focus on usability across mobile and desktop platforms directly informs MITO's cross-platform design requirements. However, a significant gap exists in the application environment and core functionality, and this is a formal one. Al-Qassab (2021) System is a specialized internal worktime management and project organization tool for a private consulting firm, relying on the Microsoft PowerApps ecosystem. The project, MITO, is a Ticketing and Task Management System for a community partner municipal office, which requires a broader, more public-facing function. The key gaps are the need to integrate public service requests (ticketing), a function absent in Al-Qassab internal worktime tracker, and to address the unique complexities of public accountability, citizen communication, and adherence to government-specific workflows, which extend beyond the scope of a commercial consulting firm's internal project management.

2.4 Synthesis of the Review

The review of related literature provides a robust framework for the development of a comprehensive ticketing and task management system by synthesizing key findings from web, desktop, and mobile application development. The collective research validates the necessity of a multi-platform solution that is not only functional and efficient but also intelligent, scalable, and deeply aligned with user behavior and organizational needs. This synthesis integrates the new research into the existing framework, reviewing the foundational and advanced precedents for the MITO: Ticketing and Task Management System, grouped by application domain: Web,

Desktop, and Mobile.

The body of research on **Web Application Development** provides the core architectural and functional blueprint for ticketing and task platforms. Studies by Gumarang et al. (2023) and Wonohardjo et al. (2022) Validate the necessity of a unified, web-based platform with distinct role-based access controls for simplifying service requests, moving away from manual and platform-dependent systems. Mission (2021) reinforced this by successfully demonstrating a centralized, multi-channel ticket management interface for organizational efficiency. Hampf (2021) adds a vital technical blueprint by confirming the viability and cross-platform benefits of developing a Progressive Web Application (PWA) for task management, which directly informs MITO's mobile and desktop usability requirements. Furthermore, Prastama et al. (2023) introduced a critical user experience (UX) framework, stressing that even robust systems require rigorous usability analysis. Jain et al. (2025) presented a forward-looking model, validating the efficiency of microservices architecture and intelligent clustering. However, a consistent gap across these web precedents is their focus on internal technical support or IT service management (ITSM) within universities or private firms, and in the case of Hampf (2021), a simple, single-user plaintext file format. This fails to address the unique complexities of public-facing governmental service, multi-user relational data, administrative workflow, and public accountability required by the MITO system.

Research in **Desktop Application Development** provides essential guidance on enhancing administrative efficiency and operational oversight. Lindblom (2024) and K. D. Aglibar & Rodelas (2022) Both highlight that maximizing system potential requires integrating the ticketing tool with other systems, strategic automation, and a strong focus on user-friendliness to mitigate the impact of high-volume tickets on staff productivity. K. D. M. Aglibar et al. (2023) Further affirms the utility of a consolidated system over manual tracking, providing a strong conceptual and functional blueprint for the "MITO: Ticketing and Task Management System" by validating core features essential for an effective platform, such as collaboration, reporting, and efficient issue resolution. Al-Hawari & Barham (2021) contribute a crucial design element by successfully demonstrating a system that integrates machine learning for automatic

classification and KPI measurement, which directly informs MITO's goal of optimizing staff workflows and improving service delivery. Meanwhile, Gayathri et al. (2022) and Montecillo et al. (2024) offer technical and conceptual precedents, respectively: the former by demonstrating successful implementation of cloud-based security and priority-based optimization, and the latter by validating user acceptance of combined task management apps to manage workload. Theoretically, Gupta et al. (2022) and Huang et al. (2024) underscore the critical need for sophisticated task scheduling and process management for efficiency. The major gap here is the context and system scope. The focus of K. D. M. Aglibar et al. (2023) on the private, commercial IT industry and an existing tool (JIRA) for bug tracking, along with the other studies' concentration on technical service desks and private sector operations, means MITO must adapt these principles of automatic classification, efficiency, and core features to human-centric administrative duties, public service requests, and non-technical complaints in a local governance setting, where the system must also manage a comprehensive Task Management component for all office duties and where accountability and public transparency are paramount.

Studies on **Mobile Application Development** offer crucial insights into human factors and advanced task planning for the municipal staff's mobile component. Ahmetoglu et al. (2024) and Zhao et al. (2023) emphasize the necessity of research-driven design that integrates strategies to mitigate human cognitive biases like the "planning fallacy" and incorporates social accountability to enhance staff productivity and user adoption. Martín-García & Pérez Marín (2020) Suggest gamification as a motivational strategy. On a conceptual frontier, Lee et al. (2024) and Song et al. (2024) present advanced models for modular task learning and sequential task planning using cutting-edge AI. Furthermore, Zhang et al. (2024), in their rigorous, state-based system evaluation framework for mobile UI task automation agents, provides an advanced conceptual validation for verifying the correct traversal of system states during complex task execution, a principle directly applicable to ensuring the reliability of MITO's ticketing and task processes. Al-Qassab (2021) provides a practical case for cross-platform (mobile and desktop) data gathering and worktime management for efficient task distribution. The fundamental gap is the domain shift. All mobile research either focuses on personal academic productivity or bleeding-edge AI mobile automation.

MITO must translate these principles of personalization, accountability, and efficient task planning from an individual or highly technical context into a reliable, professional, team-based administrative environment for public service delivery and governmental accountability.

Conclusion

In conclusion, the combined literature strongly validates the architectural necessity, functional core, and efficiency-enhancing principles required for the MITO: Ticketing and Task Management System, encompassing PWA design, task prioritization, automatic classification, and user-centric features. Successful precedents confirm the need for a unified, web-based, role-based, multi-platform system that integrates task tracking, real-time monitoring, and a user-centered design. However, a significant and pervasive research gap exists in the application domain and core purpose. Most existing studies focus on internal technical support, academic task management, or highly specialized AI/IoT environments. The MITO project is distinct because it moves the 'ticket' from an internal technical problem to a public-facing governmental service request or complaint, and the 'task' from a computational or academic duty to a human-centric administrative workflow within a municipal office. This critical shift necessitates the development of a novel system that integrates the validated principles of efficiency and usability with essential public sector requirements: external public submission, inter-departmental delegation, robust accountability mechanisms, and public transparency, functionalities that are not the focus of the current academic and commercial ticketing literature.

CHAPTER III

RESEARCH METHODOLOGY

This chapter deals with the presentation of the research design, research locale, population and sampling, research instrument, data collection procedure, and data analysis procedure.

3.1 Research Design

This study uses a quantitative research design, specifically descriptive research, which focuses on gathering measurable data from respondents to evaluate the effectiveness and usability of the MITO Ticketing and Task Management System. This design was chosen because it allows the researchers to collect numerical data and describe user perceptions objectively without manipulating variables. According to (Yoon, 2023) Descriptive quantitative research is suitable when the goal is to measure and summarize users' perceptions toward a system or application. Therefore, this design is appropriate for assessing how well the system meets the operational needs of the Municipal Information Technology Office (MITO) in Montalban, Rizal.

A purposive sampling technique was applied, selecting MITO personnel/staff, department heads, and system admin (MITO OIC or designated admin) directly involved in IT service request handling. A purposive sampling technique is appropriate in contexts where the respondents have specific expertise or involvement in the phenomenon under study. (Hossan et al., 2023; Memon et al., 2025). In this study, selecting MITO personnel directly engaged in IT service request handling ensures that the sample is information-rich and relevant to the evaluation. A total of 15 respondents were selected to ensure sufficient representation of the key personnel engaged in MITO's operations. This sample size is considered adequate for gathering meaningful insights and validating the system's effectiveness based on their direct experience and expertise.

Data was collected using a researcher-made survey questionnaire developed based on the ISO/IEC 25010 Software Quality Model. The questionnaire also included

an optional open-response section intended solely for supplementary descriptive remarks. Responses from this section were not subjected to qualitative analysis and were used only to provide additional descriptive context to the numerical survey results.

Quantitative data were analyzed using descriptive statistical techniques, including frequency, percentage, and weighted mean, to summarize the respondents' ratings. Computational processing was performed using Microsoft Excel. To ensure content validity, the questionnaire underwent expert validation by faculty members and IT professionals before administration.

The study was limited to the respondents from MITO personnel/staff, department heads, and system admin (MITO OIC or designated admin) in Montalban, Rizal. Findings were based solely on the numerical ratings provided by the selected respondents and may not be generalized to other contexts or government offices. Participation was voluntary, and informed consent was obtained from all respondents. Data privacy was strictly observed, ensuring that no personal or sensitive information was disclosed. Results were reported collectively without identifying individual respondents.

Overall, the research design aligned with the study's goal of developing and evaluating a cross-platform ticketing and task management system through quantitative assessment. The use of standardized survey instruments and statistical analysis ensured that the findings were reliable, valid, and ethically obtained, providing an objective basis for evaluating the system within the operational context of MITO.

3.2 Research Locale

The study was conducted at the Municipal Information Technology Office (MITO), under the Municipal Government of Montalban, Rizal. Montalban, officially known as Rodriguez, is a first-class municipality in the province of Rizal, Philippines. It is considered one of the fastest-growing municipalities in the region, with a population of 451,383 as of the 2024 census. (*2024 Census of Population (POPCEN)*)

Population Counts Declared Official by the President | Philippine Statistics Authority | Republic of the Philippines, n.d.). The Municipal Information Technology Office is located within the municipal hall and provides digital infrastructure, technical support, and IT services to various departments of the local government.

This setting was chosen because MITO was directly responsible for handling IT-related service requests and managing digital operations within the local government. At the time of the study, the office faced challenges due to the absence of an integrated system for handling tickets and tasks, which often led to delays, miscommunication, and inefficiencies. Since MITO managed and supported the IT infrastructure of the municipal government, it was the most appropriate setting for testing and implementing a ticketing and task management system. The proposed system aimed to address the very challenges encountered by MITO, such as inefficient task tracking, poor visibility of requests, and data privacy risks. Conducting the study in this area ensured that the developed system is both practical and contextually relevant. The study was limited to the operations of MITO and did not cover other government offices outside its direct scope. The number of respondents was limited to MITO personnel/staff, department heads, and system admin (MITO OIC or designated admin) involved in IT service requests. Access to sensitive data was restricted due to privacy concerns, and the evaluation focused primarily on system functional suitability, compatibility, interaction capability, reliability, and security.

3.3 Population and Sampling

The target population of this study consisted of the MITO personnel/staff, department heads, and system admin (MITO OIC or designated admin) within the local government of Montalban, Rizal. These individuals were directly involved in MITO's operations, particularly those engaged in IT service request handling, task management, and system usage. The study used purposive sampling, a non-probability sampling method in which respondents are deliberately selected based on their expertise and direct involvement in the phenomenon being studied. A purposive sampling technique is appropriate in contexts where the respondents have specific expertise or involvement

in the phenomenon under study. (Hossan et al., 2023; Memon et al., 2025). In this study, these groups were identified as the most appropriate respondents because they could provide relevant, experience-based feedback on the system's functional suitability, compatibility, interaction capability, reliability, and security.

Since this study used purposive sampling, no statistical formula was applied to determine the sample size. Instead, the researchers selected respondents based on their expertise, direct involvement in IT service request handling, and familiarity with MITO's operations. As noted by Memon et al. (2025) In "Purposive Sampling: A Review and Guidelines for Quantitative Research," purposive sampling relies on the researcher's judgment in selecting participants who could best contribute to the study objectives, and sample size should be justified based on the research context and analytical requirements rather than statistical power. Following this guideline, a total of 15 participants was deemed sufficient to obtain valid and representative feedback for system validation, considering the limited number of personnel within MITO and related offices.

3.4 Research Instrument

The primary research instrument used in this study was a researcher-made survey questionnaire designed to evaluate the MITO Ticketing and Task Management System. The instrument was researcher-made, but its structure and criteria were adapted from the ISO/IEC 25010 Software Quality Model, which is a widely recognized standard for evaluating software systems. The purpose of the instrument was to measure the functional suitability, compatibility, interaction capability, reliability, and security of the developed system as perceived by its actual users.

The questionnaire was divided into sections representing the software quality characteristics of ISO/IEC 25010, namely: Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security. Each item was rated using a 4-point Likert scale (1 = Strongly Disagree to 4 = Strongly Agree) to obtain quantitative measurements of the system's quality characteristics. The instrument consisted of

approximately 25 items, evenly distributed across the five ISO 25010 quality characteristics. In addition to the scaled items, the questionnaire included an optional open-ended section that allowed respondents to provide written comments regarding the system. These responses were not subjected to qualitative analysis and were used solely to supplement the interpretation of the quantitative results without forming part of the formal data analysis.

The instrument underwent content validation by a panel of IT experts and research advisers to ensure clarity, relevance, and alignment with system objectives. Revisions from the validators were incorporated before the finalization of the instrument. The questionnaire was written in English. Respondents were assured of confidentiality and anonymity. Participation was voluntary, and informed consent was obtained prior to the administration of the instrument. Data collected was used solely for academic purposes.

3.5 Data Collection Procedure

The data gathering process for this study was conducted in a supervised and structured environment within the Municipal Information Technology Office (MITO) to ensure consistency, accuracy, and validity of results. Respondents interacted with the developed Ticketing and Task Management System using their respective designated platforms (desktop, web, or mobile). This setup ensured consistent guidance and monitoring while allowing respondents to evaluate the system according to their actual user roles. The data collection focused on obtaining quantitative responses from MITO personnel/staff, department heads, and the system administrator (MITO OIC or designated administrator) to evaluate the performance and usability of the developed system. The primary data of the study consisted of numerical ratings obtained through a structured survey instrument aligned with predefined evaluation criteria. An open-ended section was included in the questionnaire solely to allow respondents to clarify or elaborate on their numerical ratings, without being subjected to qualitative analysis.

Prior to administering the survey, the researchers secured approval from the

Officer-in-Charge of MITO and other concerned municipal authorities. The finalized survey questionnaire was reviewed by IT experts for content validity to ensure clarity and reliability. Respondents were then oriented towards the purpose of the study, their role as participants, and how to use the developed system prior to answering the questionnaire. Participants were selected through purposive sampling, which is appropriate when respondents possess specialized knowledge or experience relevant to the study's focus. (Hossan et al., 2023; Memon et al., 2025). Eligible participants included those directly engaged in IT service request handling and system use. This included MITO personnel/staff, department heads, and system admin (MITO OIC or designated admin). Invitations to participate were coursed through MITO management, ensuring that respondents understand their voluntary involvement in the study.

The data gathering was scheduled in coordination with MITO management to avoid disruption of normal operations. Once respondents were available, the system was demonstrated to them, allowing hands-on exploration of features such as ticket submission, task assignment, and tracking. After using the system, participants were asked to complete the survey questionnaire. The researchers then collected the completed surveys physically and prepared the data for analysis.

Ethical considerations were strictly observed. Informed consent was obtained from all participants, and their participation remained voluntary with the option to withdraw at any point. Anonymity and confidentiality were ensured, and all data were used solely for academic and research purposes. Potential challenges, such as limited respondent availability due to workload, were addressed through flexible scheduling and continuous coordination with MITO management to ensure full cooperation and data integrity.

3.6 Data Analysis Procedure

The study utilized a researcher-made survey questionnaire based on the ISO/IEC 25010 Software Quality Model to collect quantitative data for system evaluation. The survey employed a 4-point Likert Scale ranging from Strongly Disagree (1) to Strongly Agree (4) to obtain numerical ratings from respondents regarding the system's quality characteristics.

To analyze the collected data, descriptive statistics were utilized, specifically frequency and percentage distribution to show the proportion of responses, and weighted means to determine the overall evaluation of the MITO Ticketing and Task Management System across the defined software quality attributes. These attributes included Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security. According to Starbuck (2023) Descriptive statistics are vital in quantitative studies as they help organize, summarize, and present numerical data collected through surveys in a clear and meaningful way. Data encoding and tabulation were performed using Microsoft Excel.

The data analysis followed a systematic process in which survey responses were collected, encoded, tabulated, and computed for frequency, percentage, and weighted mean using the following formulas:

Frequency (f): Refers to the number of respondents who selected each response option. Frequency counts are fundamental in descriptive statistics as they summarize how often each category or score occurs in a dataset. (*What Is: Frequency Count - Understanding Data Analysis*, n.d.).

$$f = n$$

where:

f = frequency of responses

n = number of respondents choosing a particular option

Percentage (%): Percentage represents the proportion of observations in a particular category relative to the total number of observations, allowing researchers to

summarize categorical data and show distribution trends across responses (Kotronoulas et al., 2023). Shows the proportion of responses for each option relative to the total number of responses.

$$Percentage = \frac{f}{N} \times 100$$

where:

f = frequency of a specific response

N = total number of respondents

Weighted Mean (WM): Determines the overall assessment of each statement based on the Likert scale. The weighted mean is commonly used in the analysis of Likert-type scale data to represent the central tendency of responses, particularly when the scale is treated as having approximately equal intervals between points. While some methodological discussions highlight ongoing debates regarding the use of means for ordinal data, many researchers still adopt the weighted mean for summarizing and interpreting survey results due to its ability to reflect variations in response intensity. (Koo & Yang, 2025; Yaska & Nuhu, 2024).

$$Weighted\ Mean = \frac{\sum(f \times w)}{N}$$

where:

f = frequency of responses per scale value

w = weighted value (1–4)

N = total number of respondents

The computed weighted mean was then interpreted using the descriptive scale shown in Table 1, which provides the verbal interpretation for each level of agreement based on ISO/IEC 25010. This interpretation guided the evaluation of whether the developed system meets user expectations in terms of software quality attributes.

Interval: The interval in a Likert scale represents the numerical range or distance between response categories, such as “Strongly Disagree,” “Disagree,” “Agree,” and “Strongly Agree.” According to Roselidyawaty & Rokeman (2024) defining these intervals ensures that each response option is treated as equally spaced, allowing for

valid quantitative analysis of ordinal data. This assumption enables researchers to apply statistical measures such as the weighted mean and interpret responses in a consistent, comparable manner. Roselidyawaty & Rokeman further emphasizes that clearly defining and maintaining equal intervals between scale points improves the accuracy, reliability, and interpretability of survey results. Properly structured intervals prevent bias in respondents' perception of scale distances and ensure that descriptive statistics, such as means and percentages, truly reflect participants' attitudes or evaluations.

The interval for each scale category was determined using the formula:

$$Interval = \frac{Highest\ scale\ value - Lowest\ scale\ value}{Number\ of\ levels}$$

Substituting the values:

$$Interval = \frac{4 - 1}{4} = 0.75$$

This means that each descriptive category spans an interval of 0.75. It is important to note that while the theoretical interval is 0.75 for each category, the succeeding ranges (e.g., 1.76–2.50, 2.51–3.25, 3.26–4.00) appear slightly shorter (0.74) due to rounding adjustments made to prevent overlap between categories. This ensures that each weighted mean value corresponds to only one verbal interpretation, maintaining clarity and accuracy in the analysis (Koo & Yang, 2025; Rakha et al., 2022).

Likert Scale: The Likert scale, developed by Rensis Likert, remains one of the most reliable tools in measuring subjective attitudes and perceptions in quantitative form. According to Huh & Gim (2025), the Likert scale, while often seen as ordinal, could also be treated as a continuous variable when the response options are evenly spaced. This means researchers could use averages and other statistical methods to analyze the data more accurately. Their study explains that treating the scale this way helps provide clearer and more reliable results, especially when evaluating system quality and user feedback.

CHAPTER IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presented, analyzed, and interpreted the quantitative data gathered from IT experts and system users. It included the results of the IT experts' validation to determine whether the system met its objectives and was ready for deployment, as well as the evaluation results from respondents who assessed the system's quality. The presentation of data was divided into two parts: (1) validation of the developed system by IT experts, and (2) system evaluation by respondents.

4.1 IT Experts' Validation of the System

This section presented the results of the system validation conducted by selected IT experts. The purpose of the validation was to determine whether the system met its intended objectives and was ready for deployment. The evaluation was conducted using a 4-point Likert scale ranging from "Strongly Disagree" (1) to "Strongly Agree" (4). A total of three (3) IT experts from the Institute of Computing Studies were selected to validate the developed system. They evaluated the system in terms of functional suitability, compatibility, interaction capability, reliability, and security, based on their professional perception of the system's overall performance and quality.

To assess whether the system effectively performed its intended functions and delivered accurate results, the IT experts evaluated the system in terms of Functional Suitability. This criterion measured how well the developed system provided the necessary features and functions to meet user requirements and accomplish designated tasks. The results of the evaluation were presented in Table 2.

Table 2 showed the IT experts' validation results for the system's Functional Suitability, including the weighted means, verbal interpretations, and ranks for each statement. As shown in the table, the highest-rated items were "The system provides all essential functions to meet users' tasks and goals" and "The system's functions effectively help users achieve specific tasks", both receiving a weighted mean of 3.33, interpreted as "Strongly Agree", and ranked 1.5. These results fell under the description

“The system fully meets this requirement”, indicating that the system effectively performed its intended functions and supported users in completing specific tasks.

The statements “The system produces accurate results for each function” and “The system fully covers all functional requirements across different use cases” both received a weighted mean of 3.00, interpreted as “Agree”, and ranked 3.5. These fall within the description “The system meets this requirement to a satisfactory extent”, showing that the system adequately met these functional requirements.

The lowest-rated statement, “The system’s functionality meets both explicit and inferred user needs”, obtained a weighted mean of 2.67, interpreted as “Agree”, and ranked 5th. Although it still falls within the “Agree” range, its lower mean value suggested that there was room for improvement in addressing inferred or implicit user needs that may not have been explicitly considered during system design.

Overall, the system’s functional suitability achieved an overall weighted mean of 3.07, interpreted as “Agree”, which belongs to the description “The system meets this requirement to a satisfactory extent”. This indicated that the system successfully met the objectives of the project: the web application allowed department heads to manage tickets, assign tasks, track progress, and generate reports efficiently; the mobile application enabled municipal staff to update assigned tasks, attach files, and coordinate in real time; and the desktop application provided administrators with centralized control over ticket management, user and department oversight, analytics, security, and system settings. Collectively, these findings confirmed that the developed system performed the essential operations required by the Municipal Information Technology Office and was functionally suitable for deployment.

Following this, the next criterion assessed was Compatibility, which focused on the system’s ability to operate efficiently across different environments and interact smoothly with other platforms or systems. It ensured that the developed system could integrate and function properly without conflicts or performance issues. The findings on this criterion were shown in Table 3.

Table 3 showed the IT experts' validation results for the system's Compatibility, including weighted means, verbal interpretations, and ranks for each statement. As shown in the table, the highest-rated statement, "The system operates effectively alongside other software in a shared environment", obtained a weighted mean of 3.67, interpreted as "Strongly Agree", and ranked 1st. This indicated that the system fully met the requirement for operating effectively with other applications, ensuring smooth performance in shared environments.

The next statement, "The system seamlessly exchanges and uses data with other systems," received a weighted mean of 3.33, also interpreted as "Strongly Agree", and ranked 2nd. This suggested that the system fully meet the requirement for data exchange and interoperability with other systems.

Meanwhile, "The system shares resources with minimal impact on other software" and "The system integrates data from commonly used external systems" both received a weighted mean of 3.00, interpreted as "Agree", and ranked 3.5. According to the rating scale, this indicated that the system met these requirements to a satisfactory extent, although minor refinements could further optimize its resource-sharing and data integration capabilities.

Lastly, the lowest-rated statement, "The system supports standardized formats or APIs that enhance interoperability," received a weighted mean of 2.67, interpreted as "Agree", and ranked 5th. Based on the scale, this implied that the system met this requirement to a satisfactory extent but could be enhanced by adopting or improving compliance with standardized formats and APIs for broader interoperability.

Overall, the system's compatibility received an overall weighted mean of 3.13, interpreted as "Agree", which indicated that the system met compatibility requirements to a satisfactory extent. However, continuous improvements in standardization and resource sharing were recommended to ensure smooth integration across different software environments.

The Compatibility results confirmed that the system effectively supported the study's objectives by enabling the web application for department heads, the mobile application for municipal staff, and the desktop application for administrators to operate across platforms efficiently, which ensured smooth data exchange, real-time updates, and centralized management.

Following this, the IT experts assessed the system's Interaction Capability (Usability), which evaluated its ease of use and overall user experience. This criterion measured how intuitive, accessible, and user-friendly the system interface was, as well as how easily users could navigate and perform tasks. The results of this assessment were presented in Table 4.

Table 4 showed the IT experts' validation results for the system's Interaction Capability (usability), including weighted means, verbal interpretations, and corresponding ranks for each statement. As shown in the table, the highest-rated statement was "Users could easily recognize that the system meets their needs", which obtained a weighted mean of 3.33, interpreted as "Strongly Agree" and ranked 1st. Based on the rating scale, this meant that the system fully met this requirement, clearly communicated its purpose, and aligned well with user expectations.

The statement "The system's design makes core tasks straightforward and intuitive" received a weighted mean of 3.00, interpreted as "Agree" and ranked 2nd. According to the description for this range, this implied that the system met this requirement to a satisfactory extent, suggesting that while core tasks were generally easy to perform, minor adjustments could have further improved the intuitiveness of the system's design.

The remaining statements "New users could quickly learn to use the system's primary functions", "The system effectively prevents or guides users to avoid common errors", and "The user interface is engaging and motivating for continued use" each obtained a weighted mean of 2.67, interpreted as "Agree" and ranked 3.5. This indicated that these aspects also met the requirements to a satisfactory extent, but could have benefited from improvements to enhance learnability, error prevention, and overall user engagement.

Overall, the system's interaction capability received an overall weighted mean of 2.87, interpreted as "Agree". According to the verbal interpretation scale, this signified that the system met this requirement to a satisfactory extent, demonstrating that it is generally user-friendly and accessible, with some areas for refinement to optimize usability and the overall user experience.

The Interaction Capability results confirmed that the system met its intended objectives across all platforms: the web application allowed department heads to manage tickets, track progress, and generate reports efficiently; the mobile application enabled staff to access and update tasks in real time, coordinate through notifications, and attach files; and the desktop application provided administrators with centralized control over system settings, user and department management, and analytics. Overall, the findings showed that the system effectively facilitated task execution while remaining intuitive and usable, with some room for refinement to enhance user experience.

Following this, the Reliability criterion was evaluated to assess the system's stability and consistency during operation. This aspect determined whether the system performed its intended functions accurately under specific conditions without failure or unexpected behavior. The IT experts' assessment of this criterion was presented in Table 5.

Table 5 showed the IT experts' validation results for the system's Reliability, including the weighted mean, verbal interpretation, and rank for each statement. As shown in the table, the statements "The system performs its functions consistently without faults under regular operation", "The system is resilient to minor hardware or software faults", and "The system effectively recovers and restores data after an interruption" shared ranked 2nd, with the highest weighted mean of 3.00, interpreted as "Agree". According to the given scale, this meant that the system met these reliability requirements to a satisfactory extent, indicating consistent performance and effective recovery after interruptions.

Meanwhile, the statements “The system is reliably available whenever users need it” and “The system prevents downtime and maintains steady performance” both ranked 4th with a slightly lower weighted mean of 2.67, also interpreted as “Agree”. Based on the description of the scale, this implied that the system met these aspects to a satisfactory extent, but there were occasional limitations in maintaining steady availability and continuous operation.

Overall, the system’s reliability obtained an overall weighted mean of 2.87, interpreted as “Agree”, signifying that the system performed reliably to a satisfactory extent. However, minor refinements could have further enhanced uptime consistency and overall performance stability.

The IT experts’ evaluation of the system’s Reliability indicated that the system consistently performed its functions, recovered effectively from interruptions, and maintained satisfactory performance stability. This ensured that the web application allowed department heads to manage tickets, tasks, and reports without disruption, the mobile application enabled staff to update tasks and coordinate in real time reliably, and the desktop application provided administrators with continuous control over ticket management, user accounts, and system monitoring.

Following this, the Security criterion was evaluated to determine the system’s capability to protect data and ensure authorized access to information. This aspect measures how effectively the system prevents unauthorized access, data breaches, and other potential security threats. The IT experts’ assessment of this criterion was presented in Table 6.

Table 6 showed the IT experts’ validation results for the system’s Security, including the weighted mean, verbal interpretation, and rank for each statement. As shown in the table, the statements “The system restricts access to authorized users only” and “The system prevents unauthorized modifications to data” both obtained the highest weighted mean of 3.33, interpreted as “Strongly Agree” and ranked 1.5. According to the scale description, this meant that the system fully met this requirement, demonstrating that it effectively limited access to authorized users and

protected data from unauthorized alterations.

Meanwhile, the statements “The system provides effective logs and traces of user actions for accountability” and “The system reliably authenticates user and system identities” both obtained a weighted mean of 3.00, interpreted as “Agree”, and ranked 3.5. Based on the scale, this indicated that the system met this requirement to a satisfactory extent, meaning it sufficiently ensured user accountability and proper authentication, though minor improvements could further enhance these features.

Overall, the system’s security received an overall weighted mean of 3.17, interpreted as “Agree”, this suggested that the system met its security requirements to a satisfactory extent, providing adequate data protection and user access control while leaving room for further strengthening of security measures.

Based on the IT experts’ validation, the developed system effectively met its intended objectives across all platforms. The web application allowed department heads to securely manage tickets, assign tasks, track progress, oversee employees, and generate reports and analytics. The mobile application enabled municipal staff to access and update tasks in real time, coordinate efficiently, and maintain accountability through authentication and activity logs. Meanwhile, the desktop application provided administrators with centralized control over ticketing, user and department management, system settings, monitoring, and analytics, while ensuring proper security and data protection. Overall, the system’s security features, as validated by IT experts, confirmed that it was reliable, secure, and capable of supporting efficient ticket and task management.

After evaluating the system across various quality attributes, namely Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security, the IT experts provided an overall assessment of the system’s performance and quality. This comprehensive evaluation reflects the general effectiveness of the developed system in meeting its intended objectives and readiness for deployment. The summarized results of the IT experts’ assessment for each attribute were presented in Table 7.

Table 7 showed the IT experts' validation results on the overall system quality, including the weighted mean, verbal interpretation, and rank for each software quality attribute. As shown in the table, the highest-rated attribute was Security, ranked 1st with a weighted mean of 3.17, interpreted as "Agree", meaning the system met this requirement to a satisfactory extent. This implied that the system adequately ensured data protection and prevented unauthorized access.

The attribute Compatibility, which ranked 2nd, received a weighted mean of 3.13, also interpreted as "Agree", meaning the system met these requirements to a satisfactory extent. This indicated that the system could operate effectively across different devices and environments. Functional Suitability ranked 3rd, followed with a weighted mean of 3.07, interpreted as "Agree", indicating that the system met these requirements to a satisfactory extent. It demonstrated that the system generally meets user requirements and performs its intended functions.

Both Interaction Capability and Reliability were ranked 4.5, with the lowest weighted means of 2.87, interpreted as "Agree", meaning the system met these requirements to a satisfactory extent. However, this suggested that improvements could still be made to enhance smooth interaction and system stability under various operating conditions.

Overall, the system obtained a general weighted mean of 3.02, interpreted as "Agree", which indicated that, according to the IT experts, the system met the overall software quality requirements to a satisfactory extent. Overall, the findings indicated that the developed system effectively fulfills its intended objectives and demonstrated readiness for deployment. However, the experts also suggested that minor enhancements, particularly in interaction capability and reliability, could further improve user experience and system performance.

After the system was validated by IT experts, an evaluation was also conducted among the intended users to further assess the system's effectiveness based on their actual experiences. This phase focused on gathering feedback from respondents

through a structured survey questionnaire guided by the ISO/IEC 25010 Software Quality Model. The evaluation was conducted to determine the system's level of effectiveness in terms of various quality attributes such as Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security. The results of this evaluation were presented in the succeeding tables and discussions.

4.2 System Evaluation by Respondents

This section presented the results of the respondents' evaluation of the system based on the ISO/IEC 25010 Software Quality Model, which includes Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security. The evaluation was conducted using a 4-point Likert scale ranging from "Strongly Disagree" (1) to "Strongly Agree" (4). The respondents included administrative staff, department heads, and IT personnel who tested the system's features and performance. The researchers surveyed a total of 15 respondents from MITO and MPIO offices about their perception of the developed system.

To assess the system's Functionality, the evaluation focused on how well it provides the required features and performs its intended operations accurately and effectively. This aspect examined whether the system met users' needs, produced correct outputs, and supported the completion of specific tasks. The respondents' evaluation of the system's functionality, in terms of its ability to deliver the necessary features and achieve its intended purpose, were presented in Table 8.

Table 8 showed the respondents' evaluation of the system's Functional Suitability, including the weighted mean, verbal interpretation, and rank for each statement. Based on the established rating scale Table 1, all statements fall within the 3.26–4.00 range, which corresponds to the description "The system fully meets this requirement."

As indicated in the table, the highest-rated statements were "The system allows users to create, view, and manage service tickets without errors or missing information"

and “The system provides all essential functions needed for managing requests and tasks effectively,” both obtaining a weighted mean of 3.93, interpreted as “Strongly Agree”, and ranked 1.5. This implied that the respondents found the system highly functional and efficient in supporting task management processes without issues or missing features.

Following closely, the statements “The system performs all expected features, such as assigning tasks, updating progress, and closing requests, accurately” and “Notifications and updates are displayed correctly whenever a ticket’s status changes” both received a weighted mean of 3.87, interpreted as “Strongly Agree”, and ranked 3.5. These results indicated that users found the system dependable in executing its operational features and providing timely updates during service transactions.

Meanwhile, the statement “The system provides accurate and complete reports on ticket activities and task performance” obtained the lowest weighted mean of 3.73, interpreted as “Strongly Agree”, and ranked 5th. Despite being the lowest, it still reflected high user satisfaction, suggesting that the system effectively generates detailed and reliable reports on task and ticket management.

Overall, the Functional Suitability attribute achieved a general weighted mean of 3.87, interpreted as “Strongly Agree”, which falls within the 3.26–4.00 range. This result implied that respondents believed the system fully met its functional requirements, providing accurate, complete, and reliable features that enable efficient service ticket management and task monitoring. All statements were rated highly, showing that the system effectively supported creating, viewing, and managing tickets, performing task-related features, displaying notifications, and generating reports. These results confirmed that the system met its objectives across platforms: the web application allowed department heads to manage performance, tasks, and reports; the mobile application enabled staff to update tasks in real time and coordinate efficiently; and the desktop application provided administrators with centralized control, monitoring, and analytics. Overall, the functional suitability validated that the system performs its intended functions effectively.

Following this, the next criterion evaluated was Compatibility, which examined the system's ability to operate effectively across different environments and devices while maintaining proper integration with other systems or applications. This ensured that the system could function without conflict or disruption when interacting with various platforms or software components. The respondents' evaluation of the system's compatibility based on these criteria was presented in Table 9.

Table 9 showed the respondents' evaluation of the system's Compatibility, including the weighted mean, verbal interpretation, and rank for each statement. As presented in the table, the highest-rated statements were "The system allows data to be shared and synchronized without conflicts or loss of information" and "Updates or system enhancements do not affect compatibility with existing data or features", both obtaining a weighted mean of 3.87, interpreted as "Strongly Agree" and ranked 1.5. Based on the rating scale range of 3.26–4.00 on Table 1, this falls under the description "The system fully meets this requirement." These results indicated that the system effectively maintains data integrity and remains stable even after updates or enhancements.

The statement "The system runs properly on different devices and operating systems without issues" received a weighted mean of 3.80, also interpreted as "Strongly Agree" and ranked 3rd, suggesting that users experienced smooth and consistent system performance across various devices and platforms. Meanwhile, the statements "System features remain consistent regardless of the device used" and "The system works well with other applications used by the office, such as email or file storage" both obtained weighted means of 3.67, interpreted as "Strongly Agree" and ranked 4.5. According to the same scale range (3.26–4.00), these results also fall under the description "The system fully meets this requirement." This implied that the system ensured uniform functionality across devices and integrated effectively with other software tools.

Overall, the system's compatibility achieved a general weighted mean of 3.77, interpreted as "Strongly Agree", which likewise falls within the range of 3.26–4.00. This meant that the respondents believed the system fully met its compatibility requirements, functioning efficiently across various devices, platforms, and

applications while maintaining smooth performance and data integrity. This ensured that the web application allowed department heads to monitor performance and manage tasks without issues, the mobile application enabled staff to update tasks and coordinate in real time, and the desktop application provided administrators with seamless control over ticket management, user management, and reporting. Overall, the system's strong compatibility supported its cross-platform objectives and reliable operation for all user roles.

Following Compatibility, the next aspect evaluated was Interaction Capability, which assesses how effectively the system allows users to communicate and interact with its interface and functionalities. This criterion focuses on responsiveness, user engagement, feedback mechanisms, and the overall ease of interaction between users and the system. The respondents' evaluation of this aspect was presented in Table 10.

Table 10 showed the respondents' evaluation of the system's Interaction Capability, which includes the weighted mean, verbal interpretation, and rank for each statement. As presented in the table, the highest-rated statements were "The interface is user-friendly, allowing users to easily find and use the needed functions" and "The system helps prevent mistakes by providing confirmations or alerts before performing critical actions (e.g., deleting or closing a ticket)", both obtaining a weighted mean of 3.87, interpreted as "Strongly Agree". According to the scale (3.26–4.00), this fell under the description "The system fully meets this requirement", indicating that respondents found the system highly user-friendly and appreciated its safeguards that help prevent operational errors. These statements ranked 1st among the evaluated items.

The statements "System prompts, messages, and notifications are clear and easy to understand" and "The overall layout and design are visually organized and comfortable to use" both obtained a weighted mean of 3.80, also interpreted as "Strongly Agree", which likewise falls under the description "The system fully meets this requirement." These statements ranked 3rd, suggesting that users perceived the system as clear, well-organized, and visually comfortable, promoting an effective and smooth user experience.

Meanwhile, the statement “New users could learn how to use the system quickly without extensive training” received a slightly lower weighted mean of 3.67, interpreted as “Strongly Agree”, also under the description “The system fully meets this requirement.” Despite ranking 5th, this still shows that the system is intuitive and easy for new users to learn independently with minimal guidance.

Overall, the system’s interaction capability obtained an overall weighted mean of 3.80, interpreted as “Strongly Agree”, which falls within the 3.26–4.00 range. This meant that, based on the respondents’ evaluation, the system fully met the interaction capability requirement, offering an engaging, intuitive, and user-friendly interface that supports efficient and error-free user interactions. The Interaction Capability validation confirms that the system effectively supports seamless user interaction across web, mobile, and desktop platforms, fulfilling the intended objectives of providing a functional, intuitive, and error-resilient interface for all users.

Following this, the next attribute evaluated was Reliability, which examines the system’s stability, consistency, and dependability during operation. Reliability determines whether the system could perform its intended functions accurately under specific conditions without failure or unexpected behavior. The respondents’ evaluation of this aspect was presented in Table 11.

Table 11 showed the respondents’ evaluation results for the system’s Reliability, including the weighted means, verbal interpretations, and ranks for each statement. As presented in the table, the statements “The system operates consistently without freezing or crashing during use” and “The system quickly recovers from minor technical issues without losing important data” both obtained the highest weighted mean of 3.73, verbally interpreted as “Strongly Agree”, and ranked 1.5. Based on the scale in Table 1, this value falls within the 3.26–4.00 range, which meant “the system fully meets this requirement.” This indicated that users experienced stable and dependable system performance during operation, as well as quick recovery from minor technical issues.

The statements “The system could handle multiple users performing tasks simultaneously without performance issues” and “The system provides consistent and stable performance across different sessions of use” both received a weighted mean of 3.67, interpreted as “Strongly Agree”, and ranked 3.5. According to the same scale, this range also signifies that the system fully meets this requirement, implying that respondents found the system capable of maintaining consistent performance even under multiple-user conditions and across repeated usage sessions.

Lastly, the statement “The system remains available and functional whenever users need to access it” obtained a weighted mean of 3.60, interpreted as “Strongly Agree”, and ranked 5th. Although slightly lower than the other statements, this still falls within the 3.26–4.00 range, meaning the system fully meets this requirement. This reflects users’ strong agreement that the system is accessible and functional when needed.

Overall, the system’s reliability achieved an overall weighted mean of 3.68, verbally interpreted as “Strongly Agree”, which likewise falls within the 3.26–4.00 range. This indicated that the system fully meets the reliability requirement, demonstrating stability, consistency, and dependable performance during operation, with only minor areas that could be refined to further ensure continuous availability. The Strongly Agree rating for Reliability demonstrates that the system effectively fulfills its intended objectives across all platforms, providing a stable, dependable, and consistent environment for web, mobile, and desktop users.

Following Reliability, the next attribute evaluated was Security, which focused on the system’s ability to protect data and resources from unauthorized access, misuse, or loss. It assesses how well the system ensures confidentiality, integrity, and availability of information. The respondents’ validation results for Security were presented in Table 12.

Table 12 showed the respondents’ evaluation of the system’s Security, including the weighted means, verbal interpretations, and ranks for each statement. As shown in the table, the highest-rated statement was “Only authorized users could access

the system using their own login credentials,” which obtained a weighted mean of 3.87, interpreted as “Strongly Agree” and ranked 1st. Based on the scale description (3.26–4.00), this meant the system fully meets this requirement, effectively restricting access to authorized users and maintaining strong authentication controls.

The statements “The system maintains secure authentications for all users,” “The system provides audit trails or logs for monitoring user activities,” and “The system ensures confidentiality and integrity of information across all platforms” each received a weighted mean of 3.80, also interpreted as “Strongly Agree” and ranked 3rd. These values likewise fall within the 3.26–4.00 range, indicating that the system fully meets the security requirements in these aspects. This reflected the system’s ability to uphold secure user verification, maintain effective monitoring mechanisms, and protect data integrity across various platforms.

Meanwhile, the statement “The system keeps confidential information protected from unauthorized access” obtained the lowest weighted mean of 3.73, interpreted as “Strongly Agree” and ranked 5th. Although slightly lower, it still falls within the 3.26–4.00 range, signifying that the system fully met this requirement and effectively protects sensitive information from unauthorized access.

Overall, the system’s security obtained an overall weighted mean of 3.80, verbally interpreted as “Strongly Agree”, which falls within the 3.26–4.00 range. This meant the system fully met the security requirements, ensuring that user data, authentication processes, and system resources are well-protected from unauthorized access and potential threats. The Strongly Agree ratings confirmed that the system effectively achieves its security-related objectives across all platforms, ensuring that data were protected and accessible only to authorized users, and that all actions were traceable through audit trails.

After evaluating the system across various quality attributes, namely Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security, the respondents provided an overall assessment of the system’s performance and quality. This comprehensive evaluation reflects the general effectiveness of the developed

system in meeting its intended objectives. The summarized results of the respondents' assessment for each attribute were presented in Table 13.

Table 13 showed the respondents' validation results on the overall system quality, including the weighted mean, verbal interpretation, and rank for each software quality attribute. The results were interpreted based on the rating scale in Table 1, where a mean score ranging from 3.26 to 4.00 corresponds to "Strongly Agree," described as "The system fully meets this requirement."

As indicated in the table, the highest-rated attribute was Functional Suitability, with a weighted mean of 3.87, interpreted as "Strongly Agree", and ranked 1st. This falls within the 3.26–4.00 range, signifying that the system fully met the functional requirements by effectively providing the necessary features to achieve users' goals and performing its intended tasks efficiently.

The attributes Interaction Capability and Security both obtained a weighted mean of 3.80, interpreted as "Strongly Agree", and ranked 2.5. These results showed that the system enabled smooth user interaction and ensured data protection against unauthorized access. Compatibility followed with a weighted mean of 3.77, also interpreted as "Strongly Agree", and ranked 4th, showing that the system operated effectively across various devices and environments without performance issues.

The lowest-rated attribute was Reliability, with a weighted mean of 3.68, interpreted as "Strongly Agree", and ranked 5th. Although it received the lowest rank, it still fell within the Strongly Agree range, meaning the system consistently performed its functions without failure during regular operation.

Overall, the system achieved a general weighted mean of 3.78, interpreted as "Strongly Agree", meaning the system fully met the software quality requirements. This signified that, according to the respondents, the developed system was highly functional, secure, and reliable, demonstrating excellent performance.

4.3 Respondents' Descriptive Feedback and Suggested Improvements

Following the quantitative evaluation of the system across its key software quality attributes, the study also analyzed respondents' open-ended responses that were collected as supplementary data to the survey instrument. These responses were used solely to contextualize and support the numerical findings obtained from the structured questionnaire and did not constitute a separate qualitative method. The summarized responses were presented in this section to reinforce the interpretation of the quantitative results and to identify areas for potential system enhancement.

Based on the aggregated responses, respondents generally acknowledged the system's effectiveness in supporting office-based ticketing and task management processes. Several respondents indicated that the system was well-developed, accounted for potential errors, and contributed to improved task and request handling. Statements such as "the ticketing and task management system made by the students was truly impressive and helpful" and "it really feels that every possible variable for error was considered" were treated as supporting observations that aligned with the high ratings obtained for functionality, reliability, and usability criteria.

In addition, respondents provided suggestions that were summarized and categorized to identify common improvement areas. These included the possible integration of a calendar of activities and electronic signature features to further enhance operational efficiency. Other responses emphasized the importance of maintaining stable system performance under varying internet conditions. General remarks expressing approval of the system and its potential applicability in a municipal setting were likewise noted as corroborative evidence of the system's practical value.

Overall, the summarized responses were consistent with the quantitative results, indicating that the system was perceived as useful and applicable in its intended environment. These supporting observations reinforce the survey findings and serve as reference points for future system refinements.

4.4 Summary of Findings

The study aimed to assess the developed system's overall quality and readiness for deployment through evaluations conducted by IT experts and end-user respondents. The findings were summarized as follows:

The IT experts' validation revealed that the system met its intended objectives and performed according to expectations across various software quality attributes, including Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security. As presented in Table 7 The Security attribute ranked first with a weighted mean of 3.17, followed by Compatibility (3.13) and Functional Suitability (3.07). Both Interaction Capability and Reliability received the lowest means of 2.87 and shared the same rank. The system obtained a general weighted mean of 3.02, verbally interpreted as "Agree", which falls within the range of 2.51–3.25, described as "The system meets this requirement to a satisfactory extent." These findings indicated that the system was functional, reliable, and ready for deployment, though minor enhancements, particularly in interaction capability and reliability could further optimize its performance.

Meanwhile, the respondents' evaluation resulted in a higher overall rating. As shown in Table 13, Functional Suitability ranked first with a weighted mean of 3.87, followed by Interaction Capability and Security, both obtaining a mean of 3.80. Compatibility (3.77) and Reliability (3.68) followed in the succeeding ranks. The system achieved a general weighted mean of 3.78, verbally interpreted as "Strongly Agree", which falls within the range of 3.26–4.00, meaning "The system fully meets this requirement." This implies that the respondents found the system to be highly functional, user-friendly, and secure, with strong compatibility and consistent performance across devices.

In addition to the quantitative ratings obtained from the evaluation instrument, respondents were also allowed to provide brief written comments to support their numerical assessments. These responses were used solely to supplement and contextualize the computed results and were not subjected to qualitative thematic

analysis. Overall, the responses were consistent with the numerical findings, indicating favorable perceptions of the system's effectiveness in supporting office-based operations and service delivery. Several respondents' comments reflected alignment with the high mean scores obtained in the usability, functionality, and reliability criteria, particularly in relation to ticketing and task management efficiency, error handling, and system structure. Suggestions for system improvement were also noted, including the addition of a calendar of activities, integration of electronic signatures, and optimization of system performance under varying internet conditions. These suggestions were recorded as user-identified areas for enhancement and did not affect the computed quantitative results.

Overall, the quantitative evaluation results indicated that the developed system met its intended objectives. The combination of high ratings across all quality attributes and positive feedback from users demonstrates that the system is functional, reliable, secure, and user-friendly, while also providing a foundation for future enhancements to further improve usability and performance. The succeeding chapter presented the summary, conclusions, and recommendations derived from these findings.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter presents a summary of the study, conclusions drawn from the findings, and recommendations proposed based on the results of the system evaluation. It provides an overview of how the “Development of a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal” achieved its objectives and addressed the identified problems. The presentation includes a concise summary of the study’s purpose, methodology, and major findings, followed by conclusions that reflect the system’s effectiveness and recommendations for further improvement and future development.

5.1 Summary

The study entitled “Development of a Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office in Montalban, Rizal” was conducted to design, develop, and evaluate a system that streamlines request handling, task monitoring, and monitoring processes within the Municipal Information Technology Office (MITO). The system was developed to address challenges in managing numerous technical support requests, tracking task progress, and maintaining efficient communication between IT personnel and department heads.

To achieve these objectives, the researchers adopted the Spiral Model as the Software Development Life Cycle (SDLC) approach, ensuring iterative design, continuous feedback, and risk management throughout system development. The system was evaluated by two groups of respondents: IT experts, who assessed the system’s compliance with software quality standards and readiness for deployment, and end users (department heads and IT personnel), who evaluated its usability and overall performance.

The evaluation used a researcher-made survey instrument based on the ISO/IEC 25010 Software Quality Model, covering five key attributes: Functional Suitability, Compatibility, Interaction Capability, Reliability, and Security. Data were analyzed

using descriptive statistics, particularly frequency, percentage, and weighted mean, to interpret the respondents' level of agreement on each system feature.

The respondents' validation results showed a general weighted mean of 3.02, verbally interpreted as "Agree." This indicated that the developed system satisfactorily meets the required software quality standards. Among the evaluated attributes, Security obtained the highest weighted mean (3.17), followed by Compatibility (3.13) and Functional Suitability (3.07). Interaction Capability and Reliability both received the lowest means (2.87), suggesting areas that may benefit from further improvement.

Results from the IT experts' validation revealed an overall weighted mean of 3.02, interpreted as "Agree", indicating that the system meets the required software quality standards to a satisfactory extent. Among the attributes, Security received the highest weighted mean of 3.17, followed by Compatibility (3.13) and Functional Suitability (3.07). Interaction Capability and Reliability both obtained the lowest weighted mean of 2.87, suggesting that while the system is stable and functional, further enhancement in interaction and performance consistency may be beneficial. Overall, the IT experts confirmed that the system is ready for deployment and capable of fulfilling its intended purpose.

Meanwhile, the respondents' evaluation yielded an overall weighted mean of 3.78, interpreted as "Strongly Agree", which falls within the range of 3.26–4.00, described as "The system fully meets this requirement." The highest-rated attribute was Functional Suitability (3.87), followed by Interaction Capability and Security, both obtained (3.80), and Compatibility (3.77). The lowest-rated attribute, Reliability (3.68), still received a "Strongly Agree" rating, indicating consistent system performance. These results demonstrated that the respondents perceived the system as highly functional, secure, and user-friendly, meeting all key quality standards.

In addition to the computed quantitative results, descriptive responses from respondents highlighted the system's strengths and potential improvements. The frequency of responses indicated positive perceptions regarding the system's practicality, security, and overall usability, reflecting consistency with the obtained weighted means for the evaluated criteria. Several respondents also identified suggested

system features such as a calendar of activities, electronic signature functionality, and the need for stable internet connectivity. These responses were treated as supplementary observations and were used solely to contextualize the quantitative results rather than to form an independent qualitative analysis.

Overall, the consolidated evaluation results from both IT experts and respondents indicated that the developed system met its stated objectives and complied with the ISO/IEC 25010 quality standards. The favorable ratings across all evaluated quality attributes demonstrated that the system was functional, reliable, and effective. Based on the quantitative findings, the system was deemed suitable as an efficient solution for managing service requests and tracking tasks within the Municipal Information Technology Office.

5.2 Conclusion

Based on the findings of the study, it was concluded that the “Cross-Platform Ticketing and Task Management System” achieved its objectives in improving request handling, task monitoring, and communication efficiency within the Municipal Information Technology Office of Montalban, Rizal, as evidenced by the computed weighted means and corresponding verbal interpretations.

The IT experts’ validation results indicated that the system complied with the established technical and functional requirements, obtaining an overall interpretation of “Agree.” This result signified that, based on the rating scale, the system met its intended purpose to a satisfactory extent. Similarly, the respondents’ evaluation yielded an overall verbal interpretation of “Strongly Agree,” indicating that users rated the system as fully meeting their operational requirements according to the defined evaluation criteria.

Additional inputs provided by respondents were recorded as supplementary observations and were consistent with the numerical evaluation results. These inputs reflected areas perceived as strengths of the system and identified possible

enhancements; however, they did not affect the quantitative ratings or overall interpretation derived from the statistical analysis.

The computed results further indicated that the system contributed to improved workflow efficiency through automated ticket creation, task assignment, and progress tracking, as reflected in the high mean scores obtained across usability, reliability, and functional performance indicators. The system's cross-platform implementation also supported accessibility for different user roles, while the applied security measures met the required standards based on expert evaluation.

In conclusion, the quantitative results confirmed that the developed system was functional, reliable, secure, and usable, as demonstrated by the obtained weighted means and verbal interpretations. The system effectively fulfilled its intended objectives and may be considered a viable solution for managing IT-related requests and tasks within the Municipal Information Technology Office.

5.3 Recommendations

Based on the findings and conclusions of the study, the researchers propose the following recommendations to enhance the functionality, usability, and overall quality of the developed Cross-Platform Ticketing and Task Management System for the Municipal Information Technology Office (MITO):

1) System Feature Enhancements

Integrate additional functionalities such as a calendar of activities to facilitate scheduling and monitoring of tasks. Implement electronic signature capability to enable secure approvals and verification of completed tasks.

2) System Performance and Accessibility

Improve internet connectivity support or enable offline features to allow uninterrupted access, especially for mobile users in areas with unstable network connections. Optimize system performance to maintain speed and reliability even when handling multiple simultaneous requests or high volumes of tickets.

3) Scalability and Integration

Consider future integration with other municipal systems or databases to support comprehensive service management across departments. Ensure that the system remains scalable to accommodate increasing numbers of users and service requests.

4) Further Development

Future researchers may expand this study by integrating advanced technologies, such as AI-based task prioritization, generation, or mobile offline functionality, to further improve system performance and usability.

By implementing these recommendations, the MITO Ticketing and Task Management System could further improve operational efficiency, enhance user satisfaction, and provide a secure and reliable platform for managing departmental tasks and requests.

5.4 Limitations of the Study

The study was subject to several limitations and operational constraints beyond the control of the researchers. First, the availability of respondents and the community partner occasionally affected scheduling, survey administration, and coordination meetings, due to their busy work schedules. Second, there were limited studies comparable to this project, which made identifying related literature for reference more challenging.

Technical challenges were also encountered during system development and deployment, particularly in ensuring consistent functionality across multiple platforms and devices (desktop, web, and mobile). Additionally, the collection and organization of required data from the office took longer than initially anticipated. Financial constraints also limit the researchers' ability to cover daily transportation and printing costs for necessary documents.

Potential server or network interruptions during system testing may have influenced users' interactions with the system, while variations in system access and performance across different platforms could have affected individual user experiences and ratings.

Despite these limitations, the quantitative data collection process using the standardized survey instrument was completed successfully, and the primary findings and conclusions remain based on the structured numerical evaluation of the system's performance and usability.